

Industrial Dissolved Oxygen Meter HD-480

Instruction Manual

CODE:M500770(I1002574000)

Preface

This manual describes the operation of the Industrial Dissolved Oxygen Meter, HD-480.

Be sure to read this manual before using the product to ensure proper and safe operation of the instrument.

Also safely store the manual so it is readily available whenever necessary.

Product specifications and appearance, as well as the contents of this manual are subject to change without notice.

■ Warranty and Responsibility

HORIBA Advanced Techno warrants that the Product shall be free from defects in material and workmanship and agrees to repair or replace free of charge, at HORIBA Advanced Techno's option, any malfunctioned or damaged Product attributable to HORIBA Advanced Techno's responsibility for a period of one (1) year from the delivery unless otherwise agreed with a written agreement. In any one of the following cases, none of the warranties set forth herein shall be extended ;

- Any malfunction or damage attributable to improper operation
- Any malfunction attributable to repair or modification by any person not authorized by HORIBA Advanced Techno
- Any malfunction or damage attributable to the use in an environment not specified in this manual
- Any malfunction or damage attributable to violation of the instructions in this manual or operations in the manner not specified in this manual
- Any malfunction or damage attributable to any cause or causes beyond the reasonable control of HORIBA Advanced Techno such as natural disasters
- Any deterioration in appearance attributable to corrosion, rust, and so on
- Replacement of consumables

HORIBA ADVANCED TECHN SHALL NOT BE LIABLE FOR ANY DAMAGES RESULTING FROM ANY MALFUNCTIONS OF THE PRODUCT, ANY ERASURE OF DATA, OR ANY OTHER USES OF THE PRODUCT.

■ Trademarks

Generally, company names and brand names are either registered trademarks or trademarks of the respective companies.

Conformable Directive



This equipment conforms to the following directives and standards;

Directives: The EMC Directives 89/336/EEC in accordance with Article 10 (1) of the Directive.

The Low Voltage Directive 73/23/EEC

Standard: [The EMC Directive] EN61326: 1997 + A1: 1998 + A2: 2001
(Emission: Class A, Immunity: for Industry location)

[The Low Voltage Directive] EN61010-1: 2001

This equipment shall not be used in the residential, commercial and light-industrial environment.

Note

If the sensor cable, transmission cable, or junction cable is extended to 30 m or longer, this product does not conform with the surge immunity specified in the EMC directives.

FCC rules

■ WARNING

This equipment has been tested and found to comply with the limits a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and radiates radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Installation Environment

- Pollution Degree2

Safety Policy

● Meaning of signal words

- Warning:** This indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
- Caution:** This indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.
- Mandatory:** This indicates actions you are required to do. The actual content of requirement is described using figures and text along with pictographs.
- Prohibited:** This indicates actions you are prohibited to do. The actual content of prohibition is described using figures and text along with pictographs.

■ Notational conventions used in this manual

Notes and cautions are described in the following styles in this manual:



Warning

The internal solution of DO sensor is strong alkaline. In case that the diaphragm of DO sensor is broken and the internal solution is attached to your clothes or skin, wash it off thoroughly.
Since the cornea of eye is vulnerable to alkali, pay extreme attention not to let the internal solution gets into your eyes. In case that it gets into your eyes, wash it out immediately.



Caution

When the instrument is turned off, the C-NC contact is shorted. Be careful when connecting a load.



Mandatory

To connect a load beyond the contact capacity or use inductive load (such as a motor, or pump), be sure to use a power relay whose ratio is higher than the load.



Prohibited

Do not operate the keys or push the panel surface with a sharp object, such as the fingernail.

■ Glossary

For explaining the operation of the converter, the following terms are being used.

Term	Explanation
Long press	To hold down the object until the lamp lights or the display changes.
Single press	To lightly press the object once.
Flash	To flicker quickly several times, and the setting is established.

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1 Overview

● Outline

- IP65 (dust and water proof) panel.
- Selectable sub-display items, including current temperature.
- Concentrated control keys on the front panel for all settings.
- A rich set of maintenance features
- Automatic validation of the sensor characteristics are acceptable at the time of calibration
- Free transmission output range
- Free voltage power supply (100 V to 240 V AC, 50/60 Hz)
- Memory backup

● Features

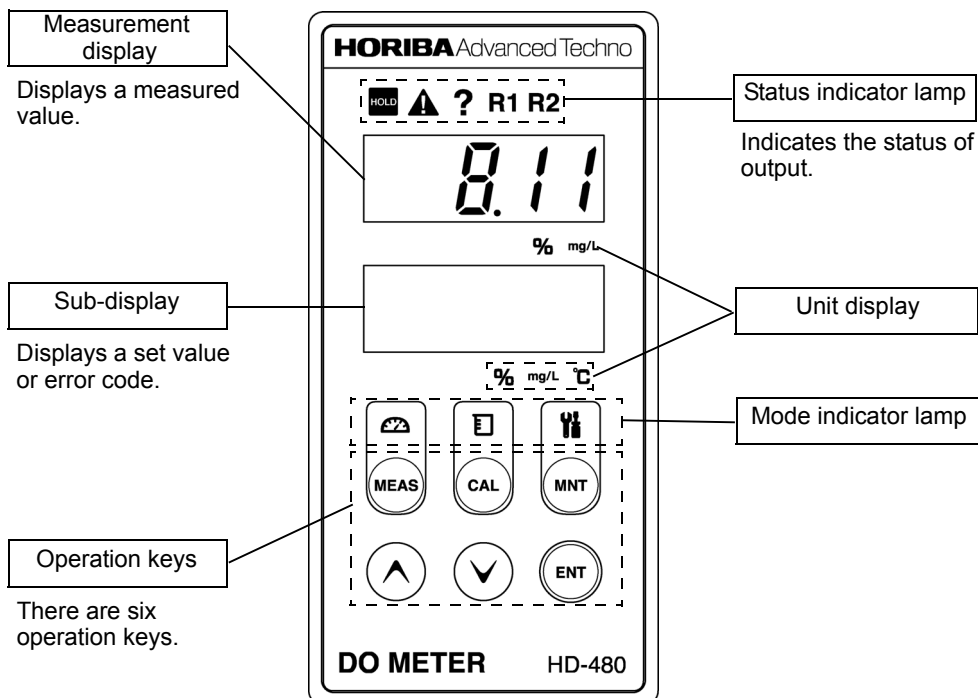
- Easy-to-read display (large characters)
- Improvement in operability of the keys using the embossed sheet
- Relaxed resistance limit of the loads to the transmission output (Max. 900 Ω)
(250 Ω receiver $\times$ 3 + wiring resistance)

● Special features

- Improved and expanded status display with icons.
- Downsized converter.
The volume is reduced by 20% compared to its predecessor.

1.1 Components

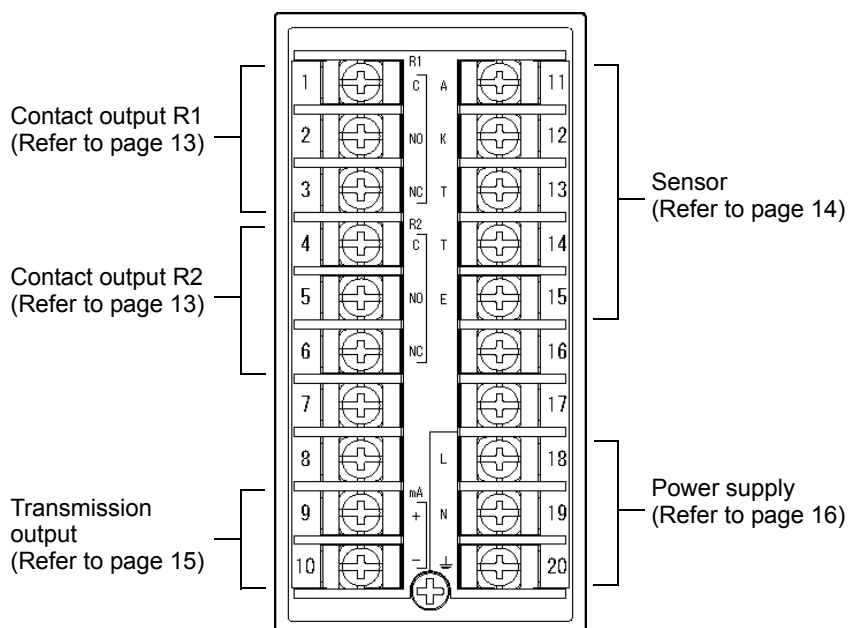
● Front panel



Prohibited

Do not operate the keys or push the panel surface with a sharp object, such as the fingernail.

● Terminal block



1.1.1 Operation keys

Operation keys are used to switch displays, enter settings, and perform calibration, etc.

You can change the value or item while the display is blinking.

Select a value or item using the UP/DOWN keys and press the ENT key. And the display will flash, and the setting will be established.

To disabled the key operations use the security menu.

Key	Notations in the text (Meaning)	Description/Operation	
	MEAS key (Measurement)	Use this key to return to the measurement mode from another mode. In the setup menus of the measurement and maintenance modes, pressing once will cancel the last change and make the display return to the previous item. To return to the measurement mode, press this key repeatedly until the MEAS lamp lights up.	
	CAL key (Calibration)	Long press	Hold down this key until the CAL lamp lights up, and the calibration mode will be enabled.
	MNT key (Maintenance)	Single press	The control value setup menu will be enabled, and "Set" will be displayed.
		Long press	Hold down this key until the MNT lamp lights up and the MEAS lamp turns off, and the maintenance mode will be enabled.
 	UP key DOWN key (Selection)	Use these keys to change the displayed value or item. For a numerical value, pressing once will increment/decrement the value by one count, and holding down the key will increase/decrease the value continuously. The UP key and the DOWN key scroll the display in the opposite directions. When you have pressed one of the buttons excessively, press the other key, to go back.	
	ENT key (Enter)	Use this key to establish set values and calibration values. Note If you use the MEAS key instead of the ENT key to return to the previous menu, the last changes are cancelled.	

1.1.2 Indicator lamps

● Mode display

	Measurement lamp Lights up during the measurement mode. This lamp is turned OFF during the calibration and maintenance modes, during which measurement is stopped.
	Calibration lamp Lights up during the calibration mode. To enter the calibration mode, hold down the CAL key until this lamp lights up. Calibration can be performed while this lamp lights up is being lit. Measurement is stopped in this state.
	Maintenance lamp Lights up during the maintenance mode. To enter the maintenance mode, hold down the MNT key until this lamp lights up. Setups can be made while this lamp lights up. Measurement is stopped in this state.

● Status display

	Lights up when the transmission output is held constant.
	Lights up when an alarm (FAIL) has been issued.
	Lights up when an error has occurred.
	Lights up when the relevant contact output is ON (C-NO is conducted).

1.2 Operation modes and menus

This instrument has 3 operation modes and 5 menus in its major category.

● Operation modes

Measurement mode

Measurements and instrument control are performed.

Calibration mode

Calibration are performed.

Maintenance mode

Measurements and outputs are stopped, and performs various setups are performed.

● Menu

The following menus are provided under the measurement and maintenance mode.
For how to enter each of the modes and menus, see the respective description pages.

Basic setup menu

This menu allows you to set up all the parameters related to measurement, such as terminal allocation of detector information.

Calibration menu

This menu allows you to performs calibration.

User check menu

This menu allows you to check the output state or measured values, and to reset the set values to the factory settings.

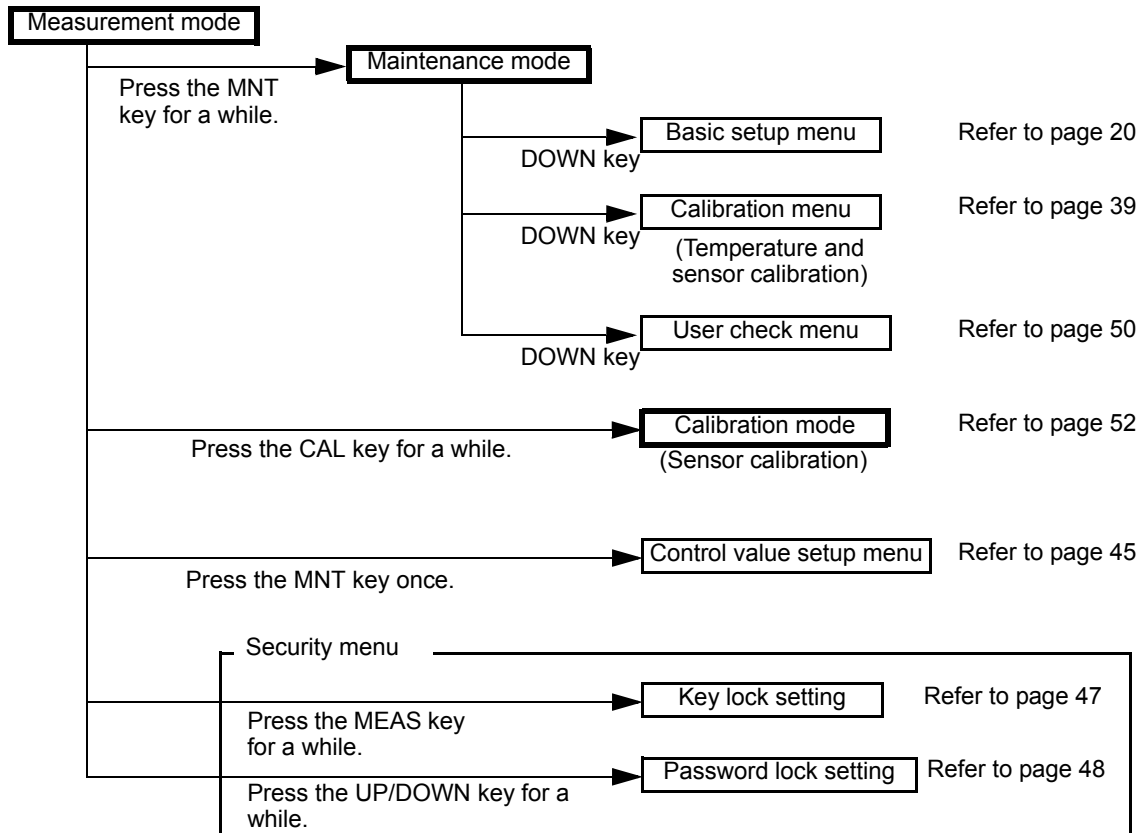
Control value setup menu

This menu allows you to set or change control values while checking the output during measurement.

Security menu

This menu prevents you from wrong or illegal operation.

● Modes and menus from the measurement mode



2 Installation

2.1 Installation environment

To keep the instrument stable and reliable for use, install it in a place where the following conditions are satisfied.

● Instrument body

- Well-ventilated
- Ambient temperature is in the range of -5°C to 45°C
- Where the air is not hot
- Not exposed to direct sunlight
- Not to exposed to direct radiant heat
- Ambient relative humidity is less than 85%
- The instrument is not splashed with water or chemicals
- Mechanical vibration is rare
- There is enough space for maintenance and wiring
- Dust and corrosive gases are not present
- The influence of electromagnetic fields is rare
- The altitude is less than 2000 m
- The range of power supply voltage fluctuation is within 10% of the rated voltage
- Overvoltage Category II is satisfied
(This regulation is applied to the electrical machinery that is powered by stationary equipment such as a switchboard.)

● Sensor

- Sensors can be checked and maintained
- No bubbles appear in the measuring solution
- Measuring solution does not corrode the wetted part of the sensor.
- The flowrate of test water is 25 cm/s or more in the sensor unit.

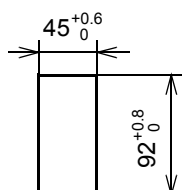
2.2 Installation procedure

2.2.1 Installing the instrument body

Mount the instrument on a control panel.

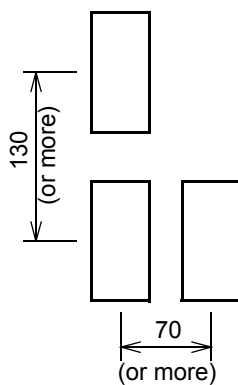
The panel thickness should be 1.0 mm to 9.0 mm.

Cut a square with dimensions shown in the figure below from the control panel.



[Unit: mm]

When installing the instruments in a row, leave an interval of 70 mm or more horizontally and 130 mm or more vertically between the square openings.

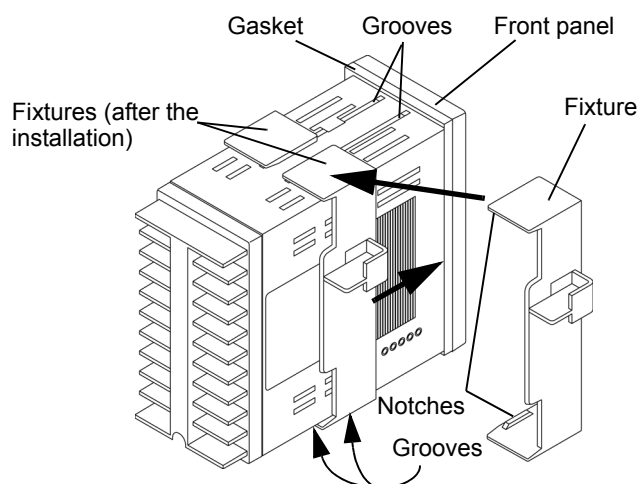


[Unit: mm]

● Mounting

Insert the instrument body into the square opening and secure it with fixtures.

1. Insert the instrument body from into the opening the front of the control panel.
2. Attach one of the two fixtures on the instrument, fitting its notches into the grooves located on the top and bottom of the body case.
3. Attach the other fixture, in the same way.
4. Slide the both fixtures to the front panel to secure the instrument body.

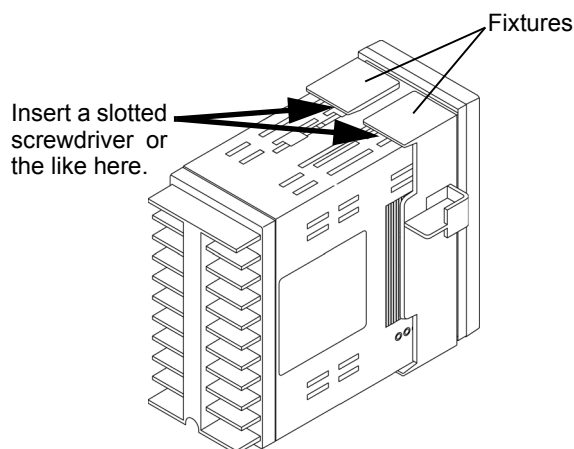


Note

The gasket is a consumable. Replace it with a new one when its holding power has become weak.

● Dismounting

1. Remove the cables from the terminal block.
Remove the fixtures one by one, and then the instrument body from the panel.
2. Insert a slotted screwdriver or the like into the gap between a fixture and the instrument body to detach a notch from the groove.
3. Remove the fixture.
4. Remove the other fixture, in the same way.
5. Pull out the instrument body from the control panel.



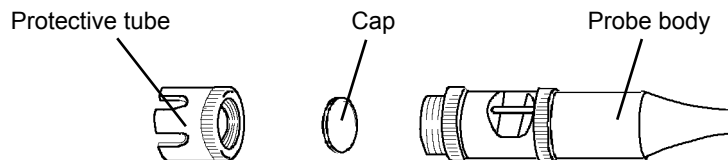
2.2.2 Installing the sensor (separately purchased)

This instrument does not come with probe and DO sensor. Please purchase it separately.

● Installing the DO sensor

Attach a DO sensor to the probe.

1. Unscrew the protective tube, and remove the cap.



Prohibited

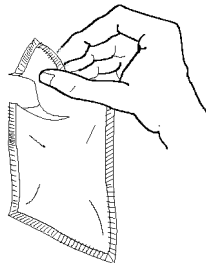
Be sure not to damage the diaphragm.
Because the diaphragm is a thin film, DO NOT hit the film with a hard object nor press it strongly. Otherwise, it may be damaged.



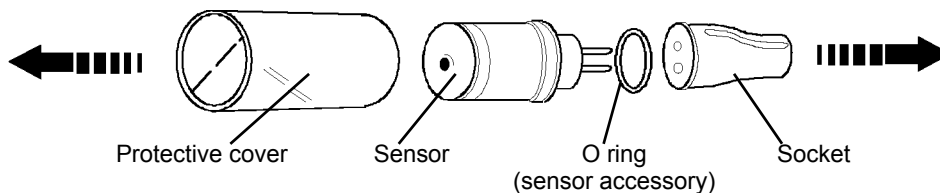
Mandatory

The internal solution of DO sensor is strong alkaline. In case that the diaphragm of DO sensor is broken and the internal solution is attached to your clothes or skin, wash it off thoroughly.
Since the cornea of eye is vulnerable to alkali, pay extreme attention not to let the internal solution gets into your eyes. In case that it gets into your eyes, wash it out immediately.

2. Take out the DO sensor from the bag.



3. Remove the tape, and then detach the protective cover and socket.



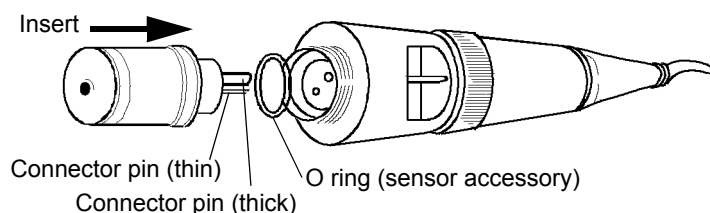
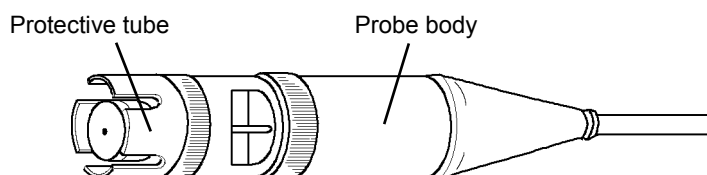
Note

Keep the bag, deoxidizer, protective cover, and socket. They will be needed when the sensor is stored.

4. Make sure that the O ring is attached to the sensor.

5. Mount the sensor to the probe.

The sensor is equipped with two connector pins, a thick one and thin one. Insert it into the probe paying attention to the direction.

**6. Put the protective tube onto the probe body and screw it firmly.**

● **Installing the sensor**

There are two methods for installing a sensor, as described below.

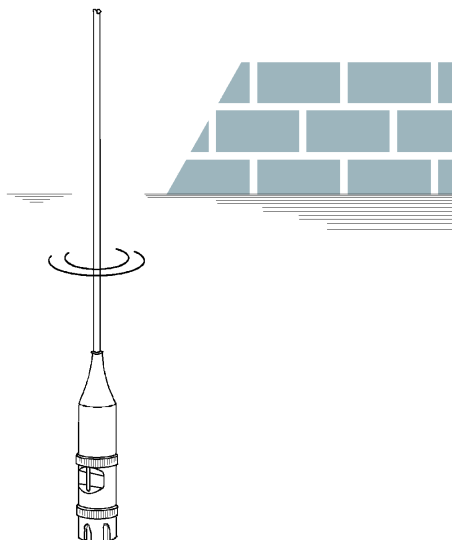
- | | |
|------------------------|--|
| Throw-in type: | The sensor is thrown into the measured solution together with the probe. |
| Immersion type holder: | The sensor is attached to the immerse type holder, and then immersed in the measured solution. |

Precautions on the sensor installation

- To measure dissolved oxygen correctly, the flowrate of 25 cm/s is required for the test water flowing in the sensor unit.
- When installed in a batch tank:
The flowrate is low near the tank wall. To make sure that the flowrate is high enough for the sensor, install the sensor in a position far enough from the wall (approx. 1 m).
Generally, if the flowrate is low, concentration gradient is generated on the surface of the sensor diaphragm, resulting in lowered readouts.
- When installed in an aerating tank
If the water current is slow, the diaphragm may attract bubbles raising the readout temporarily for the moment, resulting in unstable readout.

● **In the case of throw-in type sensor:**

- Install the probe to a place where it is immersed in the measured solution even when the level of the solution fluctuates.
- Make sure that nothing hits the sensor at the tip of the probe to prevent the film from being damaged.



● **In the case of the immersion type holder:**

Prepare a pole (50A) mounted horizontally to install the sensor.

When using the holder (DH-10-2.0) of 2 m long, 2 poles (50A) are required for installation.

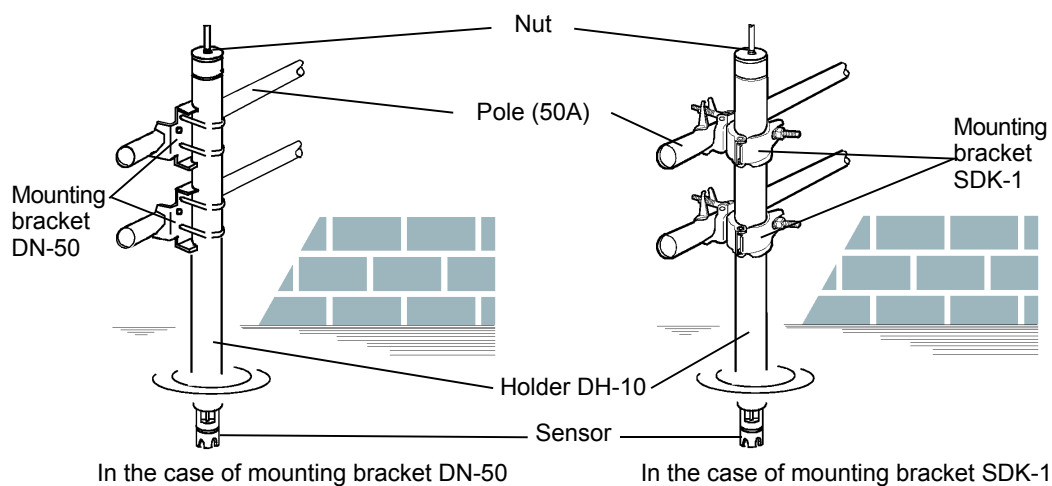
1. Attach the mounting brackets to the poles (50A) mounted in advance.

2. Secure the holder on the mounting bracket.

3. Insert the probe from the top of the holder.

Insert the probe until the tip of the probe comes out from the tip of the holder.

4. Secure the probe cable with the nut at the top of the holder.



2.3 Connection

2.3.1 Connecting the contact outputs

- The contact capacity is 240 V AC 3 A or less, or 30 V DC 3 A or less.
- If load noise is observed, use a varistor or noise killer.



Mandatory

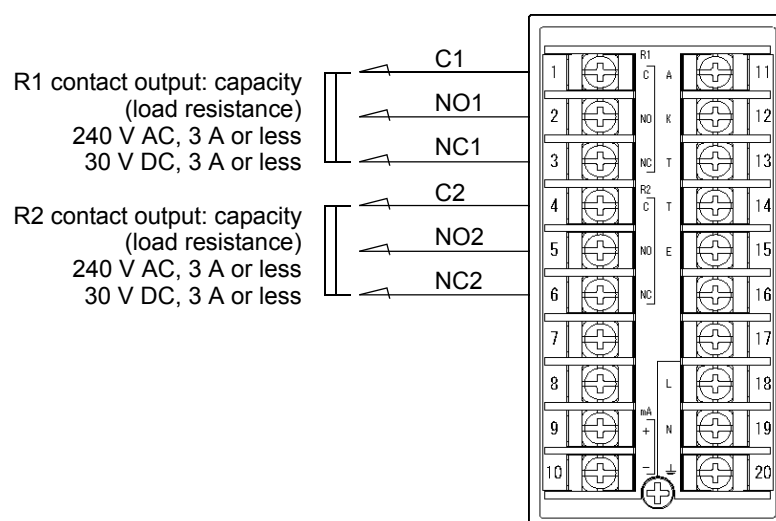
To connect a load beyond the contact capacity or use inductive load (such as a motor, or pump), be sure to use a power relay whose rating is higher than the load.



Caution

When the instrument is turned off, the C-NC contact is shorted. Be careful when connecting a load.

Make output connections as shown in the figure shown below.



Relationship between contact terminals and contact output ON/OFF

	C-NO connection	C-NC connection
Contact ON	CLOSE	OPEN
Contact OFF	OPEN	CLOSE
Power shutdown	OPEN	CLOSE

2.3.2 Connecting the sensor cable

Precautions for handling the sensor cable

The sensor cable is highly insulated. Follow the instructions below when handling it.

- Do not wet the terminals of the cable or the terminal block, nor stain them with finger marks or oil. Otherwise, the insulation of the cable is weakened.
If the insulation is weakened, the readout may become unstable. Keep the cable dry and clean all the time.
If a terminal is stained, wipe the dirt off using alcohol or the like and dry the terminal well.
- Allow enough cable length for calibration, or inspection/replacement of the sensor.
- Wire the sensor cable and junction cables keeping them off any devices that may give inductive interferences such as a motor or its cables.

● When extending the sensor cable

Use our exclusive junction cable and relay box. When ordering a junction cable, specify the length of it you need. We deliver the cables after modifying the terminals in our factory.

Reference

“7.4.1 Options” (page 63)

Precautions on extending the sensor cable

- Limit the distance from the instrument body to the sensor within 50 m.
- It is recommended to enclose the special junction cable with a conduit tube to prevent static electricity from generating due to electrical conduction, vibration, etc.

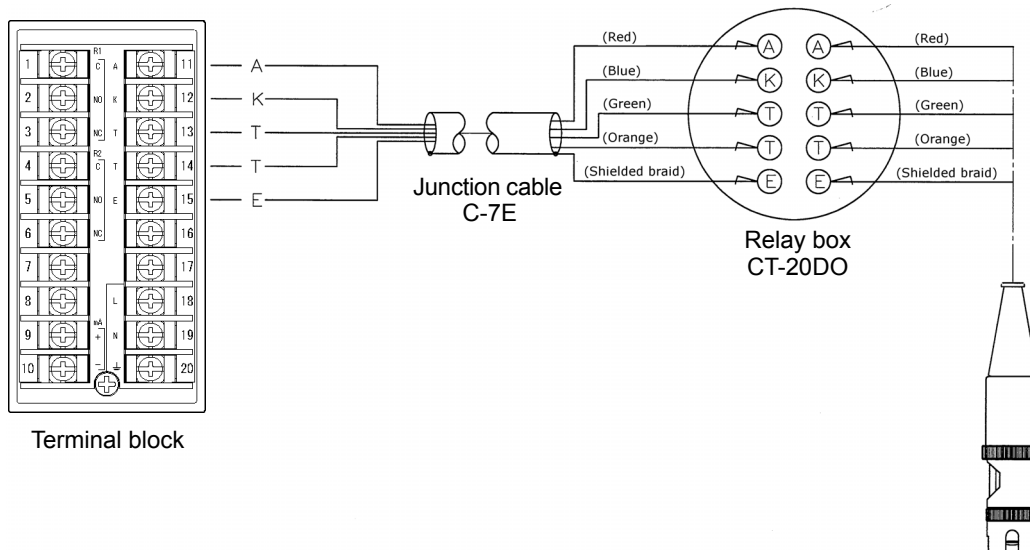
● Connections

The sensor cable is provided with the following terminals.

A	: Sensor terminal
K	: Sensor terminal
T, T	: Temperature compensation sensor terminal
E	: Shielded wire

Connect the sensor cables to the terminal block being careful to match them with the correct terminals in accordance with the figure shown below.

When you do not use a junction cable, connect the sensor cable to the terminal block directly.



Reference

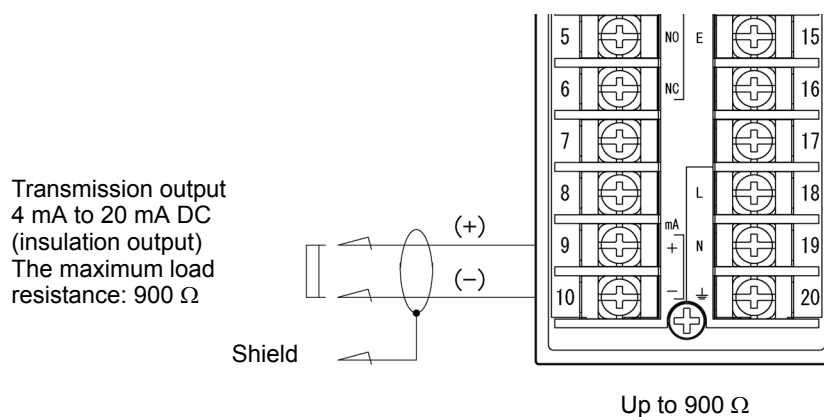
“4.2.1 Sensor type” (page 22)

2.3.3 Connecting the transmission output cable

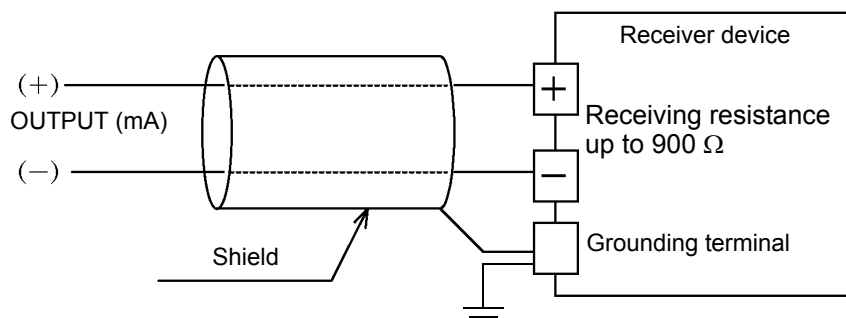
A signal of 4 mA to 20 mA DC corresponding to the measurement range is output.
The maximum allowable input resistance of the receiver device is 900 Ω .
Select an appropriate receiver device (recorder or meter relay).

1. Connect the cable to the terminal block referring to the figure below.

Use a shielded wire for the transmission cable.



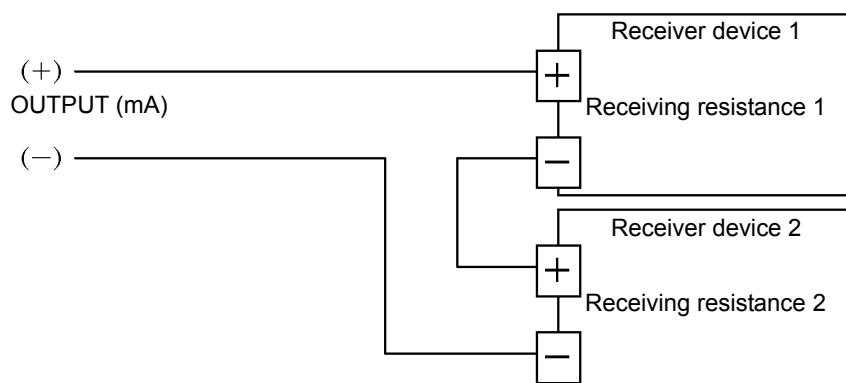
2. Ground the shielded line via the grounding terminal of the receiver device side.



● When two or more receiver devices are connected.

Connect them in a series as shown in the figure below.

The allowable total resistance of the connected receiver devices is max 900 Ω .



Mandatory

Install lightning arresters both at the instrument and receiver device.

2.3.4 Connecting the power supply



Warning

Electric Shock

Make sure that no electric power is supplied to the instrument before starting this work.

Do not turn on the power until your work is completed.

About the power switch

This instrument has no power switch. Install a power switch or circuit breaker around to turn ON/OFF the instrument.

Connecting the power cable

The power supply of this instrument is a free voltage power supply (100 V AC to 240 V AC, 50/60 Hz).

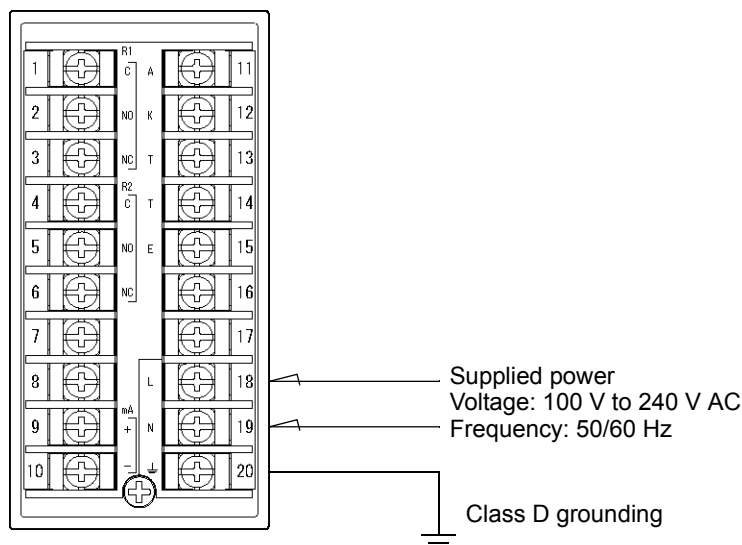
The maximum power is 10 VA.



Caution

Operating the instrument using a voltage out of the rated range may cause it to malfunction. Be sure to check the voltage of the power supply. Also make sure that the fluctuation of the power supply voltage is within the range of 10% of the rated voltage.

Connect and ground the power cable referring to the figure below.



About the grounding



Mandatory

Be sure to ground the earth terminal (class D grounding) for safety.

The grounding for the instrument should be separated from that of electrical devices such as a motor.

2.3.5 Attaching the terminal covers

Be sure to attach the terminal cover after wiring to the terminal block is completed.

3 Preparation

When you start using the instrument for the first time after factory shipment or resetting it to factory settings, be sure to complete the following preparation procedures.

- Wiring checking
- Initial set up
- Calibration

● Wiring checking

Check the following.

- Are the power cable, sensor cable, and transmission cable properly wired?
- Are the terminal block screws firmly tightened?
- Is the fluctuation of the power supply voltage within $\pm 10\%$ of the rated voltage?

● Initial set up

When the instrument is powered on in the factory setting state, it starts up in the Maintenance mode.

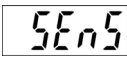
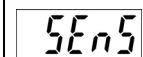
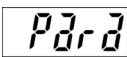
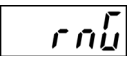
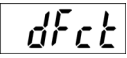
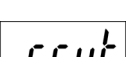
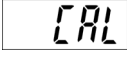
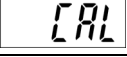
Check the current settings referring to Table 1.

If necessary, change settings following the instructions in “4.2 Basic settings” (page 20).

To finish the initial setup, press the MEAS key repeatedly until the MEAS lamp lights up to go to the measurement mode.

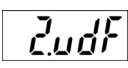
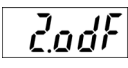
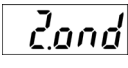
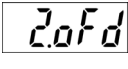
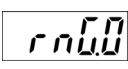
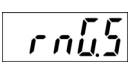
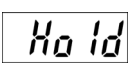
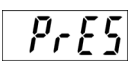
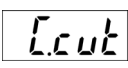
Once the initial setup is finished, the instrument starts up in the measurement mode since then. This is the normal measurement mode.

Table 1 Basic setup items

Display	Description	Options	Factory setting
	Sensor setting		
	Selection of sensor	5400, 5401, 5405	5400
	Setting of parameters used for measurement		
	Setting of the measurement items (range)	20.00 (mg/L) , 200 (%)	20.00
	Setting of times of moving average (dumping factor) 1 to 20 (times)	1 to 20 (times)	1
	Setting of salinity compensation value	0.0 to 5.0 (%)	0.0
	Display setting		
	Setting of items to appear on the sub-display	no, 1.SET, 2.SET, t, SAT.L	no
	Measurement range cut Setting not to display values outside the range specified in “Setting of the measurement items (range)”	YES, no	YES
	Calibration setting		
	Selection of calibration method	Air, SoL	Air

3 Preparation

Display	Description	Options		Factory setting
r 1 Setting for relay output 1				
1SEL	Selection of relay 1 output target	non, do, Hold, Err, FAiL, t		do
1 rE	Reverse-outputs relay 1	YES, no		no
1H-L	Selection of the operations for upper and lower limits of relay 1	H, L		H
1SEL	Setting of the control value for relay 1	DO (mg/L)	0.00 to 20.00	20.00
		DO (%)	0 to 200	200
		Temperature t (°C)	0.0 to 40.0	40.0
1dFS	Selection of the control width type of relay 1	d.diF, S.diF		d.diF
1d iF	Set values for control width in both directions	DO (mg/L)	0.00 to 4.00	0.10
		DO (%)	0 to 40	10
		Temperature t (°C)	0.0 to 10.0	2.0
1udF	For control width in a single direction: the width below the control value	DO (mg/L)	0.00 to 2.00	0.05
		DO (%)	0 to 20	5
		Temperature t (°C)	0.0 to 5.0	1.0
1odF	For control width in a single direction: the width over the control value	DO (mg/L)	0.00 to 2.00	0.05
		DO (%)	0 to 20	5
		Temperature t (°C)	0.0 to 5.0	1.0
1ond	Duration for which an alarm is output until the contact output (between C and NO) is turned ON	0 to 600 (seconds)		0
1oFd	Duration for which an alarm cancellation is output until the contact output (between C and NO) is turned OFF	0 to 600 (seconds)		0
r 2 Setting for relay output 2				
2SEL	Selection of relay 2 output target	non, do, Hold, Err, FAiL, t		do
2 rE	Reverse-outputs relay 2	YES, no		no
2H-L	Selection of the operations for upper and lower limits of relay 2	H, L		L
2SEL	Setting of the control value for relay 2	DO (mg/L)	0.00 to 20.00	0.00
		DO (%)	0 to 200	0
		Temperature t (°C)	0.0 to 40.0	0.0
2dFS	Selection of the control width type of relay 2	d.diF, S.diF		d.diF
2d iF	Set values for control width in both directions	DO (mg/L)	0.00 to 4.00	0.10
		DO (%)	0 to 40	10
		Temperature t (°C)	0.0 to 10.0	2.0

Display		Description	Options		Factory setting
		For control width in a single direction: the width below the control value	DO (mg/L)	0.00 to 2.00	0.05
			DO (%)	0 to 20	5
			Temperature t (°C)	0.0 to 5.0	1.0
		For control width in a single direction: the width over the control value	DO (mg/L)	0.00 to 2.00	0.05
			DO (%)	0 to 20	5
			Temperature t (°C)	0.0 to 5.0	1.0
		Duration for which an alarm is output until the contact output (between C and NO) is turned ON	0 to 600 (seconds)		0
		Duration for which an alarm cancellation is output until the contact output (between C and NO) is turned OFF	0 to 600 (seconds)		0
		Transmission output setting			
		Dissolved oxygen value of the zero point (4 mA) in the transmission output range	DO (mg/L)	0.00 to 22.00	0.00
			DO (%)	0 to 220	0
		Dissolved oxygen value of the span point (20 mA) in the transmission output range	DO (mg/L)	0.00 to 22.00	20.00
			DO (%)	0 to 220	200
		Setting for executing the previous value hold for the transmission output in the calibration mode (not in the maintenance mode)	YES, PrES, no		YES
		Set value for preset (optional value) hold	DO (mg/L)	0.00 to 22.00	0.00
			DO (%)	0 to 220	0
		Transmission output cut The setting for not outputting values outside the range of 4 mA to 20 mA	YES, no		YES

● Calibration

Perform the sensor and temperature calibrations referring to the following pages.

- “4.3.1 Temperature sensor calibration” (page 40)
- “4.3.2 DO sensor calibration” (page 41)

4 Measurement

4.1 Measurement

Power on the instrument with the preparation (see page 17) completed, and the instrument will start up in the measurement mode.

1. Turn the power on.

The measurement target appears on the measurement display.

The measurement range is then displayed and the instrument enters the measurement mode.

After the measurement value is displayed, and the measurement starts.

This is the normal measurement mode.

4.2 Basic settings

Enter the maintenance mode to set up the sensor type, measurement conditions, display settings, calibration method, contact output and transmission output.

● **Precautions**

- Previous value hold is used for transmission output.
- Contact outputs are the following status:
 - OFF when the control target is set to non, do, Err, or t.
 - ON when the control target is set to Hold, ON
 - FAIL when the control target is set to FAiL and a system error occurs.

● **How to open basic setup menus**

1. Hold down the MNT key in the measurement mode until the MNT lamp lights up.

The maintenance mode is enabled, and “SEt” is shown on the measurement display.

2. Press the ENT key.

The basic setup menu is enabled, and “SEnS” is shown on the sub-display.

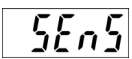
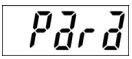
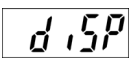
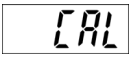
3. Press the UP/DOWN key to display the item to be set up on the sub-display.

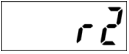
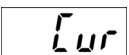
4. Press the ENT key.

The display shows the setup screen, ready for selection of setting items.

The displays shown on the sub-display and setting items are shown in the table below.

Table 2 Displays shown on the sub-display and setting items

Display	Basic setup item	Description	Reference	Remarks
	Sensor type	Sensor type	page 22	
	Measurement condition	Setting of measurement items (range)	page 22	
		Number of moving average values (dumping factor)	page 23	
		Setting of salinity compensation value	page 23	
	Display method	Sub-display setting	page 24	During measurement
		Measurement range cut ON/OFF	page 24	
	Calibration method	Switching between the air calibration and the solution calibration	page 25	

Display	Basic setup item	Description	Reference	Remarks
 	Contact output: R1/R2	Control target selection	page 27	Valid only when measured value “do” and temperature “t” have been selected for the control target.
		Reverse output setting	page 28	
		Control method setting	page 29	
		Control value setting	page 29	
		Control width type selection	page 30	
		Control width setting	page 31	
		Delay time (setting for relay ON.)	page 33	
		Delay time (setting for relay OFF.)	page 33	
	Transmission output	Transmission output setting	page 34	

Tip

Press the MEAS key to return to the menu one tier up.

To return to the basic setup menu from the setup screen, press the MEAS key repeatedly until “SEt” is shown on the measurement display and nothing is shown on the sub-display, and then press the ENT key.

4.2.1 Sensor type

Set the sensor to be used for measurements.

1. Open the sensor type setup screen referring to “● How to open basic setup menus” (page 20).

← Current setting

2. Press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to select the sensor type from 5400, (5401), and 5405.

Note

Do not select "5401" here.

4. Press the ENT key to establish the setting.

Tip

Press the MEAS key to return to the basic setup menu.

4.2.2 Measurement conditions

● Measurement items (range) setting

Set the display method of the measurement results using the range setting.

1. Open the measurement condition setting screen referring to “● How to open basic setup menus” (page 20).

The setup item selection screen is displayed on the measurement display, showing the display for measurement unit (range) setting, “rnG”.

← Current setting

2. Press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Set the measurement range by pressing the UP/DOWN keys.

20.00 : Measurement results are shown as mass of dissolved oxygen in the measured solution of 1L (mg/L). The measurement range will be 0.00 mg/L to 20.00 mg/L.

200 : Measurement results are shown as saturation of dissolved oxygen at the temperature of the measured solution (%). The measurement range will be 0% to 200%.

4. Press the ENT key to establish the setting.

Note

Even when the measurement range is set to 200, “saturation of dissolved oxygen at the temperature of the measured solution (%)”, calibration is done in “mass of dissolved oxygen in the measured solution of 1L (mg/L)”.

Tip

- To set another item of measurement setting next, press the DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Number of moving average samplings (dumping factor)

Set the number of moving average samplings used for the measurement value displayed on the screen.

1. **Open the measurement condition setting screen referring to “● How to open basic setup menus” (page 20).**

The setup item selection screen is displayed on the measurement display, showing the display for measurement unit (range) setting, “rnG”.

2. **Press the DOWN key to display “dFct” on the measurement display, and then the ENT key.**

The sub-display starts blinking. Now you can change the setting.

3. **Set the number of moving average samplings by pressing the UP/DOWN keys.**

UP key: Increments the value by 1 (time).

DOWN key: Decrements the value by 1 (time).

The settable range is 1 to 20 times.

4. **Press the ENT key to establish the setting.**

Tip

- To set another item of measurement setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Salinity compensation setting

This setting allows you to compensate for the salinity contained in the sample water. To compensate for the salinity, you must measure the salinity in the sample water in advance.

1. **Open the measurement condition setting screen referring to “● How to open basic setup menus” (page 20).**

The setup item selection screen is displayed on the measurement display, showing the display for measurement unit (range) setting, “rnG”.

2. **Press the DOWN key to display “SAL” on the measurement display, and then the ENT key.**

The sub-display starts blinking. Now you can change the setting.

3. **Set the salinity compensation value (salinity of the sample water) by pressing the UP/DOWN keys.**

UP key: Increments the value by 0.1%.

DOWN key: Decrements the value by 0.1%.

The settable range is 0.0% to 5.0%.

4. **Press the ENT key to establish the setting.**

Reference

“7.3 Saturated dissolved oxygen for temperatures and salinities (mg/L)” (page 62)

Tip

- To set another item of measurement setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

4.2.3 Display method

● Sub-display setting

Select the item to be displayed on the sub-display during measurement.

1. **Open the basic setup menu and press the UP/DOWN keys to display the current sub-display setting (as below) on the panel, referring to “● How to open basic setup menus” (page 20).**

The setup item selection screen is displayed on the measurement display, showing the display for sub-display setting, “S.dSP”.

S.dSP

no

← Current setting

2. **Press the ENT key.**

The sub-display starts blinking. Now you can change the setting.

3. **Press the UP/DOWN key to select a setting.**

no: Nothing is displayed.

1.Set: Control value of relay 1

2.Set: Control value of relay 2

t: Temperature

SAt.L: Saturation of dissolved oxygen

Note

When the measurement range has been set to 200, “saturation of dissolved oxygen at the temperature of the measured solution (%)”, “SAt.L” cannot be selected.

4. **Press the ENT key to establish the setting.**

Tip

- To set another item of current sub-display setting next, press the DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Measurement range cut ON/OFF

Select whether values outside the measurement range, are displayed or not.

1. **Open the basic setup menu and press the UP/DOWN keys to display the current sub-display setting on the panel, referring to “● How to open basic setup menus” (page 20).**

The setup item selection screen is displayed on the measurement display, showing the display for sub-display setting, “S.dSP”.

2. **Press the DOWN key to display “r.cut” on the measurement display, and press the ENT key.**

The sub-display starts blinking. Now you can change the setting.

3. **Press the UP/DOWN key to select a setting.**

YES : Values outside the measurement range are not displayed.

no : Values outside the measurement range are displayed.

4. **Press the ENT key to establish the setting.**

Tip

- To set another item of current sub-display setting next, press the UP key to display it.
- Press the MEAS key to return to the basic setup menu.

4.2.4 Calibration method

Choose a calibration method from the air calibration method and the solution calibration method.

Air calibration method

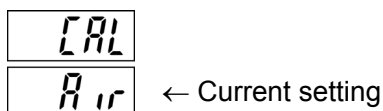
This is a simplified calibration method that calibrates the instrument by performing the zero calibration with the DO sensor removed from the probe and performing the span calibration with the sensor mounted on the probe by measuring the oxygen (O₂) in the air.

Solution calibration method

This method calibrates the instrument by using sodium sulfite water and air saturated water.

● Switching between the air calibration and the solution calibration

1. Open the calibration setting screen referring to “● How to open basic setup menus” (page 20).



2. Press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to select the calibration method.
Air: Air calibration
SoL: Solution calibration
4. Press the ENT key to establish the setting.

Tip

Press the MEAS key to return to the basic setup menu.

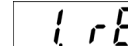
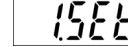
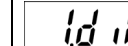
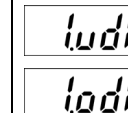
4.2.5 Contact output

Set control actions for R1 and R2 respectively.

This section explains the setup procedures using R1 as an example. Set up the R2 actions in the same way.

The following control items can be set:

Table 3 Contact output: the setting items for R1

Display	Item	Description	Reference
	Control target	Select the control target from: do: Measured value Err: Error FAiL: FAIL Hold: During maintenance t: Temperature non: No target	page 27
When the control target has been set to “FAiL”:			
	Reverse output	Select the reverse output setting, whether the output is reversed or not	page 28
When the control target has been set to “do”, measurement value, or “t”, temperature:			
	Control method	Select the limit operation, upper or lower	page 29
	Control value	Set the control value, used as the criterion for switching the relay output ON/OFF.	page 29
	Control width type	- d.diF: The same width is set for both over and under the control value at the center. - S.diF: Different widths are set for over and under the control value.	page 30
When the control width type has been set to “d.diF”:			
	Control width	Set the width (1.diF) from the control value at the center.	page 31
When the control width type has been set to “S.diF”:			
	Control width	Set the width under the control value (1.udF) and the width over the control value (1.odF).	page 31
When the contact output is used for alarms:			
	Delay time (Relay ON)	Set the alarm output duration from the start until the C-NO contact output turns ON (unit: seconds)	page 33
	Delay time (Relay OFF)	Set the alarm clear duration from the cancellation until the C-NO contact output turns OFF (unit: seconds)	page 33

● Control target selection

Select the control target of the contact output.

1. Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 (as below) on the panel, referring to “● How to open basic setup menus” (page 20).

15.61

do

← Current setting

2. Press the ENT key.

The sub-display starts blinking. Now you can change the setting.

3. Press the UP/DOWN key to select a setting.

non:	No target:	Contact output is disabled.
do:	Measured value:	Controlled by the dissolved oxygen value of the measured solution
Hold:	During maintenance:	“During maintenance” is output during calibration and maintenance
Err:	Error:	An alarm is output when any error has occurred during measurements.
FAiL:	FAIL:	An alarm is output when any failure has occurred.
t:	Temperature:	Controlled by the temperature of the measured solution

4. Press the ENT key to establish the setting.

Tip

- To set another item of relay output 1 setting next, press the DOWN key to display it.
 - Press the MEAS key to return to the basic setup menu.
-

● Reverse output setting

Select whether or not to invert the control output of the control target.

Use the reverse output for control outputs only when the control target has been set to "FAIL".

Note

When the control output is reversed, the target of the delay time setting is also reversed. When you wish to delay the FAIL output, do so by setting the delay time for relay OFF.

1. Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to "● How to open basic setup menus" (page 20).
2. Press the DOWN key to display "1. rE" on the measurement display, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to select a setting.
UP key: YES: The control output is reversed.
DOWN key: no: The control output is not reversed.
4. Press the ENT key to establish the setting.

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
 - Press the MEAS key to return to the basic setup menu.
-

● How to output FAIL during power OFF.

To output FAIL while the power supply is turned off, make the following three setting.

- Connect a cable between C and NC.
- Set the control target (1.SEL) to FAiL.
- Set the reverse output (1. rE) YES.

● Control method setting

Select the limit operation, upper or lower

1. Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).
2. Press the DOWN key to display “1.H-L” on the measurement display, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to select a setting.
UP key: H : Upper limit operation
DOWN key: L : Lower limit operation
4. Press the ENT key to establish the setting.

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Control value setting

Set the control value, used as the criterion for switching the relay output ON/OFF.

1. Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).
2. Press the DOWN key to display “1.SET” on the measurement display, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to set a setting.

The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Control target	Measurement value “do”		Temperature “t”
Measurement item (range) setting	20.00	200	
Value added by pressing the UP key	0.01 mg/L	1%	0.1°C
Value reduced by pressing the DOWN key	0.01 mg/L	1%	0.1°C
Setting range	0.00 mg/L to 20.00 mg/L	0% to 200%	0.0°C to 40.0°C

Reference

“● Control target selection” (page 27), “● Measurement items (range) setting” (page 22)

4. Press the ENT key to establish the setting.

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Control width type setting

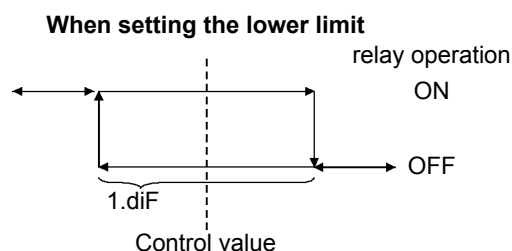
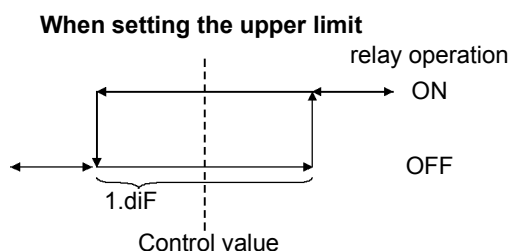
Select the control width type.

1. Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).
2. Press the UP/DOWN key to display “1.dFS”, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to select a setting.
UP key: d.diF : Sets the same width is set both over and under the control value at the center.
DOWN key: S.diF : Different widths are set over and under the control value.
4. Press the ENT key to establish the setting.

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Example of d.diF setting



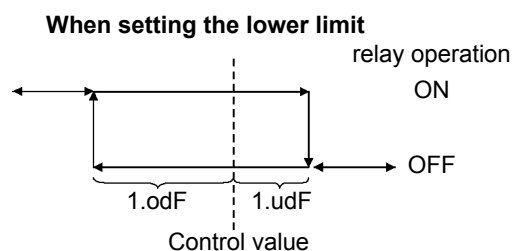
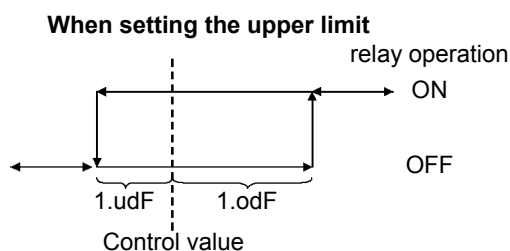
When setting the upper limit

When the control value has been set to 10.00 mg/L and 1.diF to 0.50 mg/L, the relay turns ON at 10.25 mg/L and turns OFF at 9.75 mg/L.

When setting the lower limit

When the control value has been set to 10.00 mg/L and 1.diF to 0.50 mg/L, the relay turns ON at 9.75 mg/L and turns OFF at 10.25 mg/L.

● Example of S.diF setting



When setting the upper limit

When the control value has been set to 10.00 mg/L, 1.udf to 0.50 mg/L, and 1.odf to 1.00 mg/L, the relay turns ON at 11.00 mg/L and turns OFF at 9.50 mg/L.

When setting the lower limit

When the control value has been set to 10.00 mg/L, 1.udf to 0.50 mg/L, and 1.odf to 1.00 mg/L, the relay turns ON at 9.00 mg/L and turns OFF at 10.50 mg/L.

● Control width setting

Set the control widths. The procedures are different depending on the selected control width type.

● When the control width type is “d.diF”

Sets the same width is set both over and under the control value at the center.

1. Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).

2. Press the UP/DOWN key to display “1.diF”, and then press the ENT key.

The sub-display starts blinking. Now you can change the setting.

3. Press the UP/DOWN key to select the control width.

The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Control target	Measurement value “do”		Temperature “t”
Measurement item (range) setting	20.00	200	
Value added by pressing the UP key	0.01 mg/L	1%	0.1°C
Value reduced by pressing the DOWN key	0.01 mg/L	1%	0.1°C
Setting range	0.00 mg/L to 4.00 mg/L	0% to 40%	0.0°C to 10.0°C

Reference

“● Control target selection” (page 27), “● Measurement items (range) setting” (page 22)

4. Press the ENT key to establish the setting.

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● When the control width type is “S.diF”

Different widths are set over and under the control value.

1. **Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).**
2. **Press the UP/DOWN key to display “1.udF”, and then press the ENT key.**
The sub-display starts blinking. Now you can change the setting.
3. **Press the UP/DOWN key to set the control width for under the control value.**
The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Control target	Measurement value “do”		Temperature “t”
Measurement item (range) setting	20.00	200	
Value added by pressing the UP key	0.01 mg/L	1%	0.1°C
Value reduced by pressing the DOWN key	0.01 mg/L	1%	0.1°C
Setting range	0.00 mg/L to 2.00 mg/L	0% to 20%	0.0°C to 5.0°C

Reference

“● Control target selection” (page 27), “● Measurement items (range) setting” (page 22)

4. **Press the ENT key to establish the setting.**
The display returns to the contact output setting menu.
5. **Press the DOWN key to display “1.odF”, and then press the ENT key.**
The sub-display starts blinking. Now you can change the setting.
6. **Press the UP/DOWN key to select the control width over the control value.**
The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Control target	Measurement value “do”		Temperature “t”
Measurement item (range) setting	20.00	200	
Value added by pressing the UP key	0.01 mg/L	1%	0.1°C
Value reduced by pressing the DOWN key	0.01 mg/L	1%	0.1°C
Setting range	0.00 mg/L to 2.00 mg/L	0% to 20%	0.0°C to 5.0°C

7. **Press the ENT key to establish the setting.**

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
 - Press the MEAS key to return to the basic setup menu.
-

● Delay time setting for relay ON

Set the alarm output duration from the start until the C-NO contact output turns ON (unit: seconds).

1. **Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).**
2. **Press the UP/DOWN key to display “1.ond” on the measurement display, and then press the ENT key.**
The sub-display starts blinking. Now you can change the setting.
3. **Press the UP/DOWN key to set the delay time (seconds).**
UP key: Increments the value by one second.
DOWN key: Decrements the value by one second.
The allowable value is 0 seconds to 600 seconds.
4. **Press the ENT key to establish the setting.**

Tip

- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
 - Press the MEAS key to return to the basic setup menu.
-

● Delay time setting for relay OFF

Set the alarm clear duration from the cancellation until the C-NO contact output turns OFF (unit: seconds).

1. **Open the basic setup menu and press the UP/DOWN keys to display the control target setting for R1 on the panel, referring to “● How to open basic setup menus” (page 20).**
2. **Press the DOWN key to display “1.oFd” on the measurement display, and then press the ENT key.**
The sub-display starts blinking. Now you can change the setting.
3. **Press the UP/DOWN key to set the delay time (seconds).**
UP key: Value increases in the unit for one second.
DOWN key: Value decreases in the unit for one second.
The allowable value is 0 seconds to 600 seconds.
4. **Press the ENT key to establish the setting.**

Tip

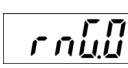
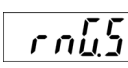
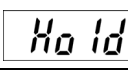
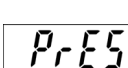
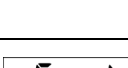
- To set another item of relay output 1 setting next, press the UP/DOWN key to display it.
 - Press the MEAS key to return to the basic setup menu.
-

4.2.6 Transmission output

Set up the items related to transmission output.

The unit and range to be used in the transmission output setting differ depending on the setting in “● Measurement items (range) setting” (page 22).

Table 4 Settings for transmission output

Display	Item	Description	Reference
	Zero point of the transmission output range	Set the measured value corresponding to 4 mA is parallelized	page 35
	Span point of the transmission output range	Set the measured value corresponding to 20 mA is parallelized	page 36
	Hold output	Select the value to be held	page 37
	Hold value	The hold value to be output when PrES, “A preset value is used as the hold value” has been selected in “● Hold output” (page 37).	page 37
	Transmission output limit	Select the transmission output setting, whether values outside the range of 4 mA to 20 mA are output or not	page 38

● Zero point of the transmission output range

Set the dissolved oxygen value corresponding to the zero point output.

1. Open the basic setup menu and press the UP/DOWN keys to display the zero setting of transmission output (as below) on the panel, referring to “● How to open basic setup menus” (page 20).

rn00

000

← Current setting

2. Press the ENT key.

The sub-display starts blinking. Now you can change the setting.

3. Press the UP/DOWN key to specify a dissolved oxygen value corresponding to the zero point output.

The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Measurement item (range) setting	20.00	200
Value added by pressing the UP key	0.01 mg/L	1%
Value reduced by pressing the DOWN key	0.01 mg/L	1%
Setting range	0.00 mg/L to 22.00 mg/L	0% to 220%

Reference

“● Measurement items (range) setting” (page 22)

4. Press the ENT key to establish the setting.

Tip

- To set another item of transmission output setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Span point of the transmission output range

Set the dissolved oxygen value corresponding to the span point output.

1. Open the basic setup menu and press the UP/DOWN keys to display the zero setting of transmission output on the panel, referring to “● How to open basic setup menus” (page 20).
2. Press the DOWN key to display “rnG.S”, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to set the dissolved oxygen value corresponding to the span point output.

The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Measurement item (range) setting	20.00	200
Value added by pressing the UP key	0.01 mg/L	1%
Value reduced by pressing the DOWN key	0.01 mg/L	1%
Setting range	0.00 to 22.00 mg/L	0 to 220%

Reference

“● Measurement items (range) setting” (page 22)

4. Press the ENT key to establish the setting.

Note

If the zero point and span point are set to the same value, the actual output is fixed to 20 mA.

Tip

- To set another item of transmission output setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Hold output

Select the hold method when the mode is switched from measurement to calibration.

Tip

When the mode is switched to maintenance, the previous value hold is used as the hold value.

1. Open the basic setup menu and press the UP/DOWN keys to display the zero setting of transmission output on the panel, referring to “● How to open basic setup menus” (page 20).
2. Press the DOWN key to display “Hold”, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to select a setting.
YES: The previous value is used as the hold value (Previous value hold).
PrES: A preset value is used as the hold value (Preset hold).
no: Hold output is not used (Actual data is output continuously).
4. Press the ENT key to establish the setting.

Tip

- To set another item of transmission output setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● Hold value

Set the hold value which is used for transmission output if the hold output has been set to PrES, “A preset value is used as the hold value”.

1. Open the basic setup menu and press the UP/DOWN keys to display the zero setting of transmission output on the panel, referring to “● How to open basic setup menus” (page 20).
2. Press the DOWN key to display “PrES” on the measurement display, and then press the ENT key.
The sub-display starts blinking. Now you can change the setting.
3. Press the UP/DOWN key to specify a dissolved oxygen value to output as the hold value.

The settings to be made differ depending on the combinations of the control targets and measurement items. See the table below:

Measurement item (range) setting	20.00	200
Value added by pressing the UP key	0.01 mg/L	1%
Value reduced by pressing the DOWN key	0.01 mg/L	1%
Setting range	0.00 to 22.00 mg/L	0 to 220%

Reference

“● Measurement items (range) setting” (page 22)

4. Press the ENT key to establish the setting.

Tip

- To set another item of transmission output setting next, press the UP/DOWN key to display it.
- Press the MEAS key to return to the basic setup menu.

● **Transmission output limit**

Select the transmission output setting, whether values outside the range of 4 mA to 20 mA are output or not.

1. **Open the basic setup menu and press the UP/DOWN keys to display the zero setting of transmission output on the panel, referring to “● How to open basic setup menus” (page 20).**
2. **Press the DOWN key to display “C.cut” on the measurement display, and then press the ENT key.**
The sub-display starts blinking. Now you can change the setting.
3. **Press the UP/DOWN key to set the transmission output limit ON/OFF.**
UP key: YES: Values under the zero point or over the span point are not output.
DOWN key: no: Values below the zero point or over the span point are output.
4. **Press the ENT key to establish the setting.**

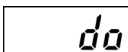
Tip

- To set another item of transmission output setting next, press the UP/DOWN key to display it.
 - Press the MEAS key to return to the basic setup menu.
-

4.3 Calibration

The display and calibration items for the calibration menu is shown below.

Perform calibration in accordance with the timing, order, and calibration items shown in the table below.

Display	Item	Timing			
		When the instrument is used for the first time	When the probe is replaced	When the DO sensor is replaced	Periodic calibration (every one month)
	Temperature calibration	1	1	-	-
	Dissolved oxygen sensor calibration				
	Zero calibration	2	2	1	-
	Span calibration	3	3	2	1

Note

The interval for the periodic calibration changes depending on the usage conditions. Check the measurement conditions regularly and perform calibrations as necessary.

● Precautions

- Wash the probe and DO sensor before performing calibration as necessary.
- Do not reuse standard solution.
- Previous value hold is used for transmission output during the maintenance mode.
- Contact outputs are in the following states during the maintenance, depending on the selected control target.
 - OFF when the control target is set to non, do, Err, or t.
 - ON when the control target is set to Hold.
 - FAIL when the control target is set to FAiL and a system error occurs.
- If the error code “E-12” (degraded sensor sensitivity) is shown during calibration, the sensor has run out of useful life and needs replacement.
- Even when the measurement range has been set to 200, “saturation of dissolved oxygen at the temperature of the measured solution (%)”, calibration is done in “mass of dissolved oxygen in the measured solution of 1L (mg/L)”.
- When a new sensor is used for the first time, it takes time for the sensor to become stable. Thus, calibration may not be possible in this state. In that case, turn on the power for the instrument with the sensor attached and wait for the sensor to become stable to perform calibration.

● How to enter the calibration mode

Enter the calibration mode from the measurement mode.

1. **Hold down the MNT key in the measurement mode until the MNT lamp lights up.**
The instrument enters the maintenance mode, and “SEt” is shown on the measurement display.
Measurement stops, and the instrument enters a state of hold.
2. **Press the DOWN key to display “CAL” on the measurement display, and then press the ENT key.**
The instrument enters the calibration menu and “t” is displayed on the sub-display.
3. **Select the calibration item you wish perform by pressing the UP/DOWN keys.**
t: Temperature sensor calibration “4.3.1 Temperature sensor calibration” (page 40)
do: DO sensor calibration “4.3.2 DO sensor calibration” (page 41)

Tip

You can open the calibration menu directly from the measurement mode. For how to do it, refer to “5.1 Daily calibration (calibration mode)” (page 52).

4.3.1 Temperature sensor calibration

Perform the calibration for the temperature sensor using two solutions whose temperatures are known and the difference between them is at least 10°C.

Tip

- To cancel the calibration in the middle of it, press the MEAS key. The instrument will return to the measurement mode without updating the calibration data.
 - If any error is displayed, take necessary actions referring to the description in “6.3 Error code” (page 57).
-

● Operation

1. **Enter the calibration menu from the measurement mode referring to “● How to enter the calibration mode” (page 40).**
2. **Immerse the sensor in the solution of the first point and take time to let the sensor to be prepared for measuring the temperature.**
3. **Press the ENT key with the instrument showing “CAL” on the measurement display and “t” on the sub-display.**
The instrument enters the temperature calibration menu, showing “t” on the measurement display and “-1-” on the sub-display.
4. **Press the ENT key.**
The temperature calibration starts for the first point.
5. **Adjust the value shown on the measurement display to match the actual temperature of the solution by pressing the UP/DOWN keys.**
UP key: Increments the value by 0.1°C.
DOWN key: Decrements the value by 0.1°C.
The settable range is -10°C to 10°C.
6. **Press the ENT key to establish the calibration for the first point.**
The sub-display shows “-2-” next, indicating that the instrument has entered the calibration for the second point.
7. **Immerse the sensor in the solution of the second point and take time to let the sensor to be prepared for measuring the temperature.**
8. **Press the ENT key.**
The temperature calibration starts for the second point.

The temperature of the second-point solution will be displayed on the measurement display in blinking.

Note

An error is caused if the temperature difference between two solutions is below 10°C.

9. After the measurement value has been stabilized, press the ENT key.

The sub-display starts blinking.

10. Adjust the value shown on the measurement display to match the actual temperature of the solution by pressing the UP/DOWN keys.

UP key: Increments the value by 0.1°C.

DOWN key: Decrements the value by 0.1°C.

The settable range is 0°C to 40°C.

11. Press the ENT key to establish the calibration for the second point.

The instrument shows “CAL” on the measurement display and “good” on the sub-display, indicating that the new temperature calibration values have been applied.

The above completes the temperature calibration.

Tip

Press the MEAS key to return to the screen for selecting calibration items.

4.3.2 DO sensor calibration

Two methods are provided for performing the DO sensor calibration, air calibration and solution calibration.

Reference

For how to switch between the two methods, refer to “4.2.4 Calibration method” (page 25).

● Air calibration

In this method, a zero calibration is performed with the DO sensor removed from the probe. A span calibration is performed with the DO sensor attached to the probe based on the oxygen (O₂) in the air.

Note

Before performing an air calibration, remove the probe from the solution and take time to let the SO sensor to dry completely.

Tip

- To cancel the calibration in the middle of it, press the MEAS key. The instrument will return to the measurement mode without updating the calibration data.
 - If any error is displayed, take necessary actions referring to the description in “6.3 Error code” (page 57).
-

● Zero calibration to span calibration

The section describes the procedure for performing an zero calibration and span calibration in succession.

1. Enter the calibration menu from the measurement mode referring to “● How to enter the calibration mode” (page 40).

-
- 2. Remove the DO sensor from the probe.**
 - 3. Press the ENT key with the instrument showing “CAL” on the measurement display and “do” on the sub-display.**
The instrument enters the calibration mode, showing “0.CAL” on the measurement display and the selected calibration method on the sub-display.
 - 4. Press the ENT key.**
The instrument shows the current measurement value on the measurement display in blinking and the temperature on the sub-display.
 - 5. Press the ENT key to establish the zero point.**
The instrument shows “S.CAL” on the measurement display and the selected calibration method on the sub-display.
Next perform a span calibration.
 - 6. Attach the DO sensor to the probe.**
 - 7. Press the ENT key.**
The instrument shows the current measurement value on the measurement display in blinking and the temperature on the sub-display.
 - 8. After the measurement value has been stabilized, press the ENT key.**
The instrument shows “CAL” on the measurement display and “good” on the sub-display, indicating that the span point has been established.

The above completes the zero and span calibrations for air calibration.

Tip

Press the MEAS key to return to the screen for selecting calibration items.

● Span calibration

This section describes the procedure for performing a span calibration alone.

- 1. Enter the calibration menu from the measurement mode referring to “● How to enter the calibration mode” (page 40).**
- 2. Press the ENT key with the instrument showing “CAL” on the measurement display and “do” on the sub-display.**
The instrument enters the calibration mode, showing “0.CAL” on the measurement display and the selected calibration method on the sub-display.
- 3. Press the DOWN key to show “S.CAL”, and press the ENT key.**
The instrument shows the current measurement value on the measurement display in blinking and the temperature on the sub-display.
- 4. After the measurement value has been stabilized, press the ENT key.**
The instrument shows “CAL” on the measurement display and “good” on the sub-display, indicating that the span point has been established.

The above completes the span calibration for air calibration.

Tip

Press the MEAS key to return to the screen for selecting calibration items.

● Solution calibration

In this method, a zero calibration is performed using a zero solution (sodium sulfite water) and a span calibration is performed using a span solution (air saturated water).

Tip

Name	Preparation
Zero solution (sodium sulfite water)	Dissolve 20 g of sodium sulfite in 100 mL of water (approx. 20%). Increase the amount of solution if necessary. The useful life of the solution is around one day.
Span solution (air saturated water)	(Example) Have a 500-mL beaker, air pump, and bubbler ready. Pour about 500 mL of water into the beaker, and bubble the water for 15 minutes to 30 minutes, and then leave it at rest for about one minute.

Precautions on solution calibration

- Flow the calibration solution at 25 cm/s or more using a magnetic stirrer or something similar.
If the flowrate is low, the span sensitivity is improperly set high.
- Set the probe so that the temperature sensor is immersed in the water fully.
- You can use saturated water containing salinity for this calibration. In this case, measure the salinity in the solution and perform salinity compensation of this instrument before performing a calibration. (Refer to “● Salinity compensation setting” (page 23).)

Tip

- To cancel the calibration in the middle of it, press the MEAS key. The instrument will return to the measurement mode without updating the calibration data.
- If any error is displayed, take necessary actions referring to the description in “6.3 Error code” (page 57).

● Zero calibration to span calibration

This section describes the procedure for performing an zero calibration and span calibration in succession.

- Enter the calibration menu from the measurement mode referring to “● How to enter the calibration mode” (page 40).
- Remove the sensor from the solution and wash it with tap water.
- Immerse the sensor in the zero solution (sodium sulfite water) and air saturated water.
- Press the ENT key with the instrument showing “CAL” on the measurement display and “do” on the sub-display.
The instrument enters the calibration mode, showing “0.CAL” on the measurement display and the selected calibration method on the sub-display.
- Press the ENT key.
The instrument shows the current measurement value on the measurement display in blinking and the temperature on the sub-display.
- Press the ENT key.
The sub-display shows “0.00” in blinking. Now you can adjust the DO value.

7. Change the value shown on the measurement display to match the DO value of the zero solution by pressing the UP/DOWN keys.

UP key: Increments the value by 0.01 mg/L.

DOWN key: Decrements the value by 0.01 mg/L.

The settable range is 0.00 mg/L to 0.30 mg/L.

8. Press the ENT key to establish the zero point.

The instrument shows "S.CAL" on the measurement display and the selected calibration method on the sub-display.

Next perform a span calibration.

9. Immerse the sensor in the air saturated water.

10. Press the ENT key with "S.CAL" being displayed.

The instrument shows the current measurement value on the measurement display in blinking and the temperature on the sub-display.

11. After the measurement value has been stabilized, press the ENT key.

The instrument shows "CAL" on the measurement display and "good" on the sub-display, indicating that the span point has been established.

The above completes the zero and span calibrations for solution calibration.

Tip

Press the MEAS key to return to the screen for selecting calibration items.

● **Span calibration**

This section describes the procedure for performing a span calibration alone.

1. Enter the calibration menu from the measurement mode referring to "● How to enter the calibration mode" (page 40).

2. Remove the sensor from the solution and wash it with tap water.

3. Immerse the sensor in the air saturated water.

4. Press the ENT key with the instrument showing "CAL" on the measurement display and "do" on the sub-display.

The instrument enters the calibration mode, showing "0.CAL" on the measurement display and the selected calibration method on the sub-display.

5. Press the DOWN key to show "S.CAL", and press the ENT key.

The instrument shows the current measurement value on the measurement display in blinking and the temperature on the sub-display.

6. After the measurement value has been stabilized, press the ENT key.

The instrument shows "CAL" on the measurement display and "good" on the sub-display, indicating that the span point has been established.

The above completes the span calibrations for solution calibration.

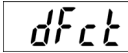
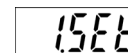
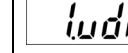
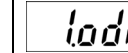
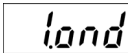
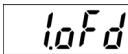
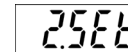
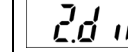
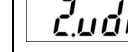
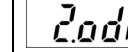
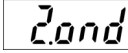
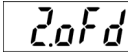
Tip

Press the MEAS key to return to the screen for selecting calibration items.

4.4 Control value setup menu

This menu allows you to change the following control-related settings while performing measurements in the measurement mode.

Table 5 Contact output: setting items

Display	Item	Description	Remarks
	Number of data used for moving average	Select the number of data used for moving average, which is performed to display measured values.	page 23
When “do” (measurement value) or “t” (temperature) has been selected in “● Control target selection” (page 27).			
	Control value	Set the control value, used as the criterion for switching the relay output ON/OFF.	page 29
When “d.diF” has been selected in “● Control width type setting” (page 30).			
	Control width	Set the control width whose center is the control value (1.diF).	page 31
When “S.diF” has been selected in “● Control width type setting” (page 30).			
 	Control width	Set the control width under the control value (1.udF) and the width over the control value (1.odF).	page 31
	Delay time (Relay ON)	Set the alarm output duration from the start until the C-NO contact output turns ON (unit: seconds)	page 33
	Delay time (Relay OFF)	Set the alarm clear duration from the cancellation until the C-NO contact output turns OFF (unit: seconds)	page 33
When “do” (measurement value) or “t” (temperature) has been selected in “● Control target selection” (page 27).			
	Control value	Set the control value, used as the criterion for switching the relay output ON/OFF.	page 29
When “d.diF” has been selected in “● Control width type setting” (page 30).			
	Control width	Set the control width whose center is the control value (2.diF).	page 31
When “S.diF” has been selected in “● Control width type setting” (page 30).			
 	Control width	Set the control width under the control value (2.udF) and the width over the control value (2.odF).	page 31
	Delay time (Relay ON)	Set the alarm output duration from the start until the C-NO contact output turns ON (unit: seconds)	page 33
	Delay time (Relay OFF)	Set the alarm clear duration from the cancellation until the C-NO contact output turns OFF (unit: seconds)	page 33

● **Operation**

1. Press the MNT key in the measurement mode.

The sub-display shows "dFct". (The measurement display keep showing a measured value.)

2. Press the UP/DOWN key to display on the sub-display.

The setting items are displayed in the order shown in Table 5.

3. Press the ENT key.

Now you can change the setting.

4. Change the setting referring to the instruction page of the selected item.

— Tip —

If you do not operate any keys for 10 seconds during item selection (step 2. mentioned above), this function is closed automatically (the display returns to the normal measurement status). During setting change (step 3. or later), this function is always open.

4.5 Security menu

This menu allows you two functions:

Key lock, to prevent accidental operation; and password lock, to prevent illegal operation.

4.5.1 Key lock

This function prevents accidental key operation, which may cause unintended setting change or hold output.

- **Set up**

1. **Hold down the MEAS key in the measurement mode displayed until “Lock” on the measurement display and “on” on the sub-display respectively.**

The key lock displayed function starts working. Releasing the key lets the instrument return to the measurement mode, and the setup for the key locking is completed.

No key operation is accepted hereafter and when any key is pressed, “Lock” is displayed on the measurement display and “on” on the sub-display respectively.

- **Release**

1. **Press the MEAS key when a measured value is displayed.**

The instrument will display “Lock” on the measurement display and “on” on the sub-display.

Note

You cannot release the key lock immediately after key lock setup, with the displays showing “Lock” and “on”. Make sure to start the release operation with a measured value displayed.

2. **Hold down the key in the measurement mode continuously until the display of the sub-display turns into “oFF” from “on”.**

When you release the MEAS key with “oFF” displayed on the sub-display, key lock release is completed.

The instrument returns to the measurement mode, and now all keys are enabled.

4.5.2 Password locking

This function prevents illegal operation by unauthorized persons using the preset first and second passwords.

Each password is a 4-digit number.

When the password lock is enabled, the instrument performs measurement and rejects any key operations except for password lock setup (mentioned below).

You can enable/disable the lock only by following the procedure described below.

Note

- Note that if the password locking is enabled, the maintenance functions such as calibration and settings are disabled. Set it up carefully.
 - The password before factory shipment has been set to 0000-0000.
-

● Password setting

You must input the current passwords before setting up new passwords.

Note

- It is recommended to use numbers which is easy to remember and hard to forget.
 - If you forget the passwords you set, send the instrument back to HORIBA Advanced Techno. We will send it back to you after resetting the passwords to the default setting (0000-0000). (Charged)
-

1. Perform the steps 1. to 5. of “Password operations” (page 49).

Measurement display shows “Lock” and the sub-display shows the current lock status, “on” for enabled or “off” for disabled.

2. Press the UP/DOWN key to select the password to be set.

3. Press the ENT key.

The instrument enters the setup state and the sub-display starts blinking.

Tip

To cancel the whole password setting and return to the measurement mode, press the MEAS key with a digit blinking.

4. Press the UP/DOWN key to set a new password.

5. Press the ENT key.

The setting is established and the digit stops blinking.

Then the next digit starts blinking.

6. Repeat steps 4. to 5. to the rightmost digit.

When the rightmost digit is established, all the four digits flashes and the password is read into the instrument.

7. To change the other password next, press the UP/DOWN key to select the other password, and change it in the same way.

● Password operations

1. **Hold down the UP key and DOWN keys at the same time in the measurement mode until "PASS" is shown on the main display.**

When you release the buttons, the measurement display shows "PAS1", and the sub-display shows "0000" with the leftmost digit blinking. You can change the value of the blinking digit.

Tip

To cancel the whole password setting and return to the measurement mode, press the MEAS key with a digit blinking.

2. **Press the UP/DOWN key to select a value for the blinking digit.**

3. **Press the ENT key.**

The setting is established and the digit stops blinking.

Then the next digit starts blinking.

4. **Repeat steps 2. to 3. to the rightmost digit.**

When the rightmost digit is established, all the four digits flashes and the first password is read into the instrument.

And then the measurement display shows "PAS2" and the sub-display shows "0000".

5. **Input the second password in the same way.**

When the two passwords are correct, the measurement display shows "Lock" and the sub-display shows the current lock status, "on" for enabled or "oFF" for disabled.

Note

If either of the passwords is wrong, the measurement display shows "PASS" and the sub-display shows "bad", and the instrument returns to the measurement mode.

The instrument does not tell you which password was wrong.

6. **If necessary, to change the setting press the ENT key.**

The instrument enters the setup state and the sub-display starts blinking.

7. **Press the UP/DOWN key to set the password locking.**

UP key: on: Enables the password lock.

Down key: oFF: Disables the password lock.

8. **Press the ENT key.**

The sub-display flashes, and the setting is established. The measurement display shows "Lock" and the sub-display shows the current lock status, "on" for enabled or "oFF" for disabled.

9. **Press the MEAS key to return to the measurement mode.**

4.6 User check menu

The user check menu allow you to check the instrument status, and to reset it to the default. Operate it from the user check menu.

● How to open the user check menu

Start the operation it from the measurement mode.

1. Hold down the MNT key until the MNT lamp lights up.

The instrument enters the maintenance mode, and “SEt” is shown on the measurement display.

2. Press the DOWN key to display “USr” on the measurement display, and then press the ENT key.

The instrument opens the user check menu, and “LEd” is shown on the sub-display.

Refer to the below instructions for perform each of the available operations in accordance with the following description.

4.6.1 Status check

You can check the four items shown below.

- LEDs
- Transmission output
- Relay operation
- Measured value and temperature

● Operations

Perform the following operations after opening the user check menu.

● LEDs

1. Press the UP/DOWN key, if necessary, to display “LEd” on the sub-display, and press the ENT key.

All the LED will light up normally.

2. Check that LEDs light up without problems.

3. Press the MEAS key.

The instrument returns to the user check menu entrance.

● Transmission output

You can output an electric current of 4 mA, 12 mA and 20 mA from the transmission output.

1. Press the UP/DOWN key to display “Cur” on the sub-display.

2. Press the ENT key.

“4-20” is shown on the measurement display and “4” on the sub-display outputting a 4 mA transmission output.

3. To switch the current output, press the UP/DOWN key to select the value corresponding to the current (mA) to be output from 4, 12, or 20.

4. Check that the electric current is being output properly.

5. Press the MEAS key.

The instrument returns to the user check menu entrance.

● Relay operation

Check the operations of contact outputs R1 and R2.

Checking the contact output R1

1. Press the UP/DOWN key to display “rLY” on the sub-display.

2. Press the ENT key.

“rLY1” is shown on the measurement display and “rLY” on the sub-display.

3. Press the ENT key.

The relay lamp R1 lights up and the relay is turned on, normally.

Press the ENT key to switch the relay ON/OFF to check the operation.

4. Press the MEAS key.

The instrument returns to the user check menu entrance.

Checking the contact output R2

You can check R2 in the same way as R1. But, press the UP/DOWN key to display “rLY2” on the sub-display after the step 2.

● Measured value and temperature

1. Press the UP/DOWN key in the user check menu to display “diSP” on the sub-display.

2. Press the ENT key.

The current measured value is shown on the measurement display and the current temperature on the sub-display.

3. Press the MEAS key.

The instrument returns to the user check menu entrance.

4.6.2 Instrument reset

This operation allows you to initialize all the settings and restart the instrument.

Then preform the steps in “● Initial set up” (page 17) in the automatically opened basic setup menu.

Calibration values are also reset. Be sure to perform the steps in “4.3 Calibration” (page 39) before use.

● Operation

1. Enter the user check menu referring to “How to open the user check menu” (page 50).

“USr” is shown on the measurement display and “LEd” on the sub-display.

2. Press the DOWN key to display “init” on the sub-display, and then press the ENT key.

The sub-display shows “no” and starts blinking.

3. Press the DOWN key to select “YES”, and then press the ENT key.

The instrument is reset to the state before factory shipment.

The sub-display shows “good” with flashing, and the instrument restarts.

Tip

- The initial settings are shown in “Table 1 Basic setup items” (page 17).
 - To return to the user check menu entrance without resetting, select “no” in step 3. and press the ENT key.
-

5 For More Accurate Measurements

This is the maintenance information to keep the precision of this instrument and to use it better.

This chapter explains the maintenance to keep the precision of the instrument such as daily calibration.

5.1 Daily calibration (calibration mode)

Here we explain how to perform daily calibration.

● Precautions for calibration

- The transmission output is put to the state set in the “● Hold output” (page 37) setting of the basic setup.
- Contact outputs are put to the following conditions:
 - Turned OFF when set to non, do, Err, or t.
 - Turned ON when set to Hold, ON.
 - FAIL is output if a system error is raised when set to FAiL.
- Even when the measurement range has been set to 200, “saturation of dissolved oxygen at the temperature of the measured solution (%)”, calibration is done in “mass of dissolved oxygen in the measured solution of 1L (mg/L)”.

● Calibration operation

Operate it from the measurement mode.

1. Hold down the CAL key until the CAL lamp lights up.

The DO sensor calibration menu appears and the calibration selected in the basic setup (air calibration or solution calibration) starts.

2. Perform a calibration in accordance with the procedure described in “● Air calibration” (page 41) or “● Solution calibration” (page 43).

Tip

- If you want to abort the calibration, press the MEAS key. The display returns to the measurement mode without updating the calibration data.
 - If any error is displayed, take necessary actions referring to the description in “6.3 Error code” (page 57).
 - To perform the temperature calibration, call up the calibration screen from the maintenance mode.
-

5.2 Maintenance for the instrument body

5.2.1 Inspection of the meter

If the insulation of the terminals for connecting the DO sensor has been weakened, accurate measurement becomes difficult.

Inspect the terminal assembly periodically (once a year or so) to check for rust and the like that cause insulation to become weak.

Wipe out the dirt from the body case with soft cloth.



Prohibited

Do not use an organic solvent or something similar.

5.3 Maintaining the DO sensor

This section describes the procedure for maintaining the DO sensors.

5.3.1 Washing the DO sensor

Stain on the DO sensor is a cause of slow response, drifting readout, and unstable readout.

Check the DO sensor periodically and wash out stains on it using tap water to keep it clean.

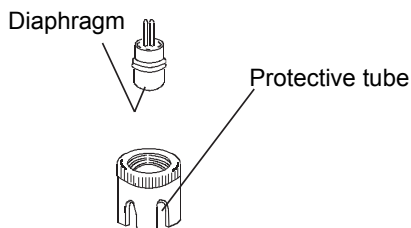
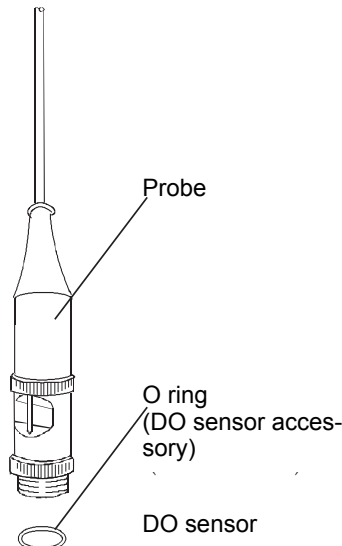
It is recommended to wash the DO sensor at the same time of calibration. In this case, be sure to perform a calibration after washing the sensor.

Note

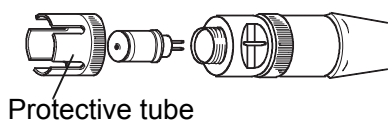
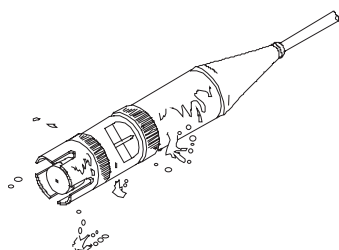
Take due care not to damage the DO sensor diaphragm.

5.3.2 Replacing the DO sensor

● Precaution on the handling



● Replacement procedure



The internal solution of DO sensor is strong alkaline. In case that the diaphragm of DO sensor is broken and the internal solution is attached to your clothes or skin, wash it off thoroughly. Since the cornea of eye is vulnerable to alkali, pay extreme attention not to let the internal solution get into your eyes. In case that it gets into your eyes, wash it out immediately.

- Be sure not to damage the diaphragm. Because the diaphragm is a thin film, DO NOT hit the film with a hard object nor press it strongly. Otherwise, it may get damaged.
- Check the O ring. An O ring is used to bridge the DO sensor and probe to prevent water from getting into the instrument. Check that the O ring is attached and secure it with the protective tube.

1. Remove the probe from the measured solution.
2. Wipe off the moisture on the probe well.

3. Unscrew the protective tube.
4. Remove the DO sensor.

Note

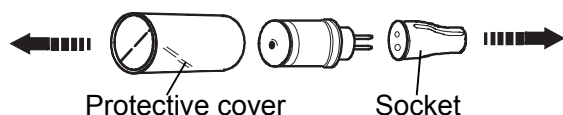
When disposing of a DO sensor, treat it as an industrial waste.

5. Make sure that the connector for the probe has no moisture in it.

6. Take out the DO sensor from the bag.



7. Remove the tape, and then remove the protective cover and socket.



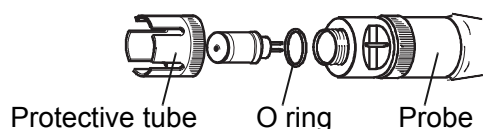
Note

Keep the bag, deoxidizer, protective cover, and socket. They will be needed when the sensor is stored.

8. Mount the DO sensor to the probe.

- The sensor is equipped with two connector pins, a thick one and thin one. Insert it into the probe paying attention to the direction.
- Make sure that the DO sensor has the O ring attached.

9. Screw the protective tube into the probe firmly.



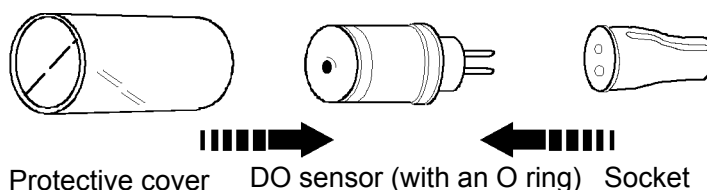
The above completes the replacement of a DO sensor.

Before making measurements, perform a temperature calibration and DO sensor calibration.

5.3.3 Storage

This section describes the procedure for removing the DO sensor from the probe and store it.

1. Remove the DO sensor from the probe referring to “5.3.2 Replacing the DO sensor” (page 54).
2. Attach the protective cover and socket. to the sensor.



3. Put the DO sensor in the bag in which the sensor was contained and a deoxidizer pack also.
4. Let the air out of the bag and completely seal the bag with heat seal, etc.
In case you have lost the bag in which the DO sensor was contained, use an aluminum-evaporated bag which does not allow oxygen to penetrate. As for deoxidizer, use the one included in the bag or a commercially available one.

Note

The storage procedure above is treatment for minimizing the degradation of the sensor.

6 Troubleshooting

6.1 Troubleshooting

Problems	Causes or possibilities	Remedies
The measurement readout is unstable.	● Bubbles stick to the sensor. ● Bubbles appear because the flow rate is too fast. ● The level of the measuring solution has changed, and thus the area of the sensor that contact with liquid has changed. ● Rapid changes in liquid temperature. ● Screws of the terminal block are loose. ● Insulation of the terminal block has been weakened.	● Do not allow bubbles to appear in the measurement liquid. ● Control the flow rate. ● Install the instrument in a way that the level of the measuring solution does not change. ● Install the instrument in a place where liquid temperature does not change drastically. ● Tighten them securely. ● Drain the moisture of the terminal block. ● Clean the terminal block.
Abnormal value is displayed.	● The sensor is stained. ● The sensor is not soaked in the sample liquid. ● Disconnection or short circuit of the sensor cable ● Loose connection with the terminal block	● Wash the sensor. ● Install the instrument in a way that the level of the measuring solution does not change. ● Check the sensor cable. ● Check the terminal block.
Response is slow.	● The flow rates is slow. ● The uneven concentration of the measuring solution (when a tank has been installed)	● Increase the flow rate. ● Stir the sample well. ● Install it in a place where the concentration is even.
The instrument cannot be turned on.	● The power supply voltage is out of the rated voltage range. ● Wiring mistake ● Internal error of the instrument	● Check the power supply voltage. ● Check the wiring. ● Contact HORIBA Advanced Techno.

6.2 Out of the measurement range

When the measurement value is out of the measurement range (-0.01 mg/L or less/-1% or less, or 20.01 mg/L or more/201% or more), "oF" is displayed on the sub-display in blinking. If that happens, follow the instructions given below.

Code	Causes or possibilities	Remedies
oF (out of the measurement range)	The concentration is too high.	Check to see if the concentration is within the measurement range by manual analysis. If it is within the measurement range, contact HORIBA Advanced Techno.
	Temperature sensor error (The temperature readout is lower than the actual temperature.)	Perform a temperature calibration.
	Terminals A and K are connected oppositely.	Check the wiring and connect terminals A and K correctly. If the instrument does not operate normally even if the terminals are correctly connected, contact HORIBA Advanced Techno.

6.3 Error code

The instrument is equipped with the function to display various error codes.
Error codes are shown on the sub-display in a blinking state.

6.3.1 Description of error codes

Error code		Error	Description	Occurrence	Contact output
E-12	Error	Span calibration error	The current for the sensor is low during span (sensitivity) calibration.	At the time of calibration ^{*1}	None ^{*1}
E-13	Error	Zero calibration error	The current for the sensor is low during zero calibration.		
E-21	FAIL	Temperature sensor disconnected	The lead of the temperature sensor is broken.	At the time of measurement and calibration	Yes ^{*2}
E-22	FAIL	Temperature sensor short-circuited	The lead of the temperature sensor has been short-circuited.		
E-23	Error	Out of the temperature compensation range	The measured temperature is 45°C or more.		
E-24	Error	Out of the temperature compensation range	The measured temperature is -5°C or less.		
E-25	Error	Temperature calibration error	The temperature calibration has failed.	System error	Yes
E-90	FAIL	System error	Initialization error		
E-91	FAIL	System error	The content of the memory such as set values and calibration values has been lost.		
E-92	FAIL	System error	Malfunction of the A/D converter		

^{*1}: The contact output is not activated even if the target of the contact output has been set to Err.

^{*2}: The contact output is not activated during calibrations.

Tip

- Errors and FAILs can be output as an anomaly by setting the basic setup items.
- If any error code is displayed, the transmission output is preset to the value just before the error happens.

6.3.2 Remedies for error codes

When an error code is displayed, take necessary actions in accordance with the table below.

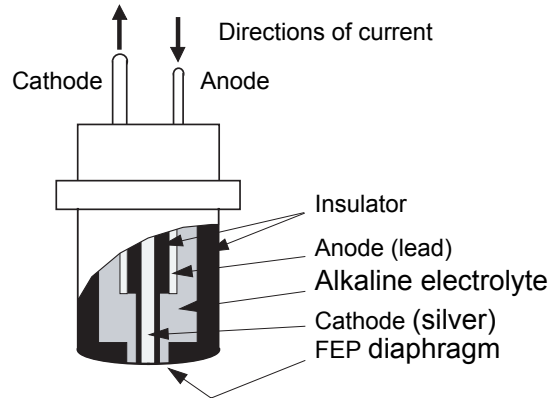
Error code	Causes or possibilities	Remedies
E-12 (span calibration error)	The sensor has desensitized.	Replace the sensor.
	The sensor diaphragm has been damaged. (It has a pin-hole, etc.)	
E-13 (zero calibration error)	When using sodium sulfite, its concentration is low.	Prepare it again.
	The calibration has not been successfully performed or the instrument has been damaged.	Recalibrate the instrument again. After the adjusted value has been stabilized, press the ENT key. If the instrument does not operate normally even after being calibrated with leads A and K disconnected, it is malfunctioning.
E-21 (temperature sensor disconnected)	Resistance error between T and T of the sensor.	If the resistance value is out of the range 6.2 k Ω to 7.7 k Ω , the sensor is defective. Replace it with a new one.
	Wrong wiring of the sensor cable or junction cable	Check to see if that the wiring is open between T and T.
E-22 (temperature sensor short-circuited)	Resistance error between T and T of the sensor.	If the resistance value is out of the range 6.2 k Ω to 7.7 k Ω , the sensor is defective. Replace it with a new one.
	Wrong wiring of the sensor cable or junction cable	Check to see if that the wiring has been shorted between T and T.
E-23 (out of the temperature compensation range)	The wiring between terminals T and T is open. (The lead wire has been disconnected.)	Check the wiring.
	The temperature sensor is defective.	Check that the resistance between T and T is within the range 6.2 k Ω to 7.7 k Ω . (If the temperature sensor is defective, replace the probe.)
	The temperature of the solution is 45°C or higher.	Lower the solution temperature to below 45°C.
E-24 (out of the temperature compensation range)	The wiring between terminals T and T has been shorted.	Replace the probe.
	The temperature sensor is defective.	Replace the probe.
	The temperature of the test water is -5°C or less.	Check the temperature of the test water.
E-25 (temperature calibration error)	The temperature sensor is defective.	Check that the resistance between T and T is within the range 6.2 k Ω to 7.7 k Ω . (If the temperature sensor is defective, replace the probe.)
	The calibration was performed when the temperature difference between the first and second points was 10°C or less.	Recalibrate the instrument using two solutions whose difference in temperature is more than 10°C.
E-90 E-91 E-92 (System error)	System error in the instrument	Turn the power OFF and turn it ON again. If the system error persists even after restarting the instrument, contact HORIBA Advanced Techno.

7 Data

7.1 Measurement principle

The structure of a diaphragm-type galvanic sensor is shown in the figure below.

In the sensor, an anode made of base metal (lead) and an cathode made of precious metal (silver) are immersed in the alkaline electrolyte partitioned by a gas permeable membrane made of FEP (fluororesin film), and the cathode is positioned closely contact with the diaphragm. This forms a galvanic cell.



The oxygen in a test water permeates the FEP film, and starts the following electrochemical reactions on the surface of the cathode.

Cathode reaction $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$

Anode reaction $2Pb \rightarrow 2Pb^{2+} + 4e^-$

The cathode surface is reduced by oxygen, causing the external load connected between the two poles to be energized. Since the current intensity changes in proportion to the partial pressure of oxygen in the test water, the dissolved oxygen can be measured by detecting this current. However, since the permeability of the diaphragm for oxygen and the air-saturated oxygen concentration in water change by temperature, the instrument detects the temperature and performs the computation for compensating for it using a microcomputer. In addition, since the concentration gradient is generated on the surface of the diaphragm if the flowrate of test water is low, the flowrate of the stipulated rate or higher is needed.

7.2 Specifications

Product name	Industrial dissolved oxygen meter		
Model	HD-480		
Measurement method	Galvanic cell method		
Measuring range	Dissolved oxygen 0 mg/L to 20 mg/L, 0.01 mg/L resolution Saturation 0% to 200%, 1% resolution Temperature 0°C to 40°C, 0.1°C resolution		
Repeatability	Within ±0.5% of the full scale (by equivalent input)		
Linearity	Within ±0.5% of the full scale (by equivalent input)		
Transmission output	4 mA to 20 mA DC, input/output isolated type Maximum load resistance: 900 Ω Transmission output range: Freely settable within the measuring range		
Contact output	Outputs: 2 points Alarm contact output (R1 and R2) Contact type: relay contact, SPDT (1c) Contact rating: 240 V AC, 3 A and 30 V DC, 3 A (resistance load) Contact function: Selectable from “Upper limit operation”, “lower limit operation” (ON/OFF control), “anomaly alarm”, “during maintenance”, and “temperature”		
Calibration function	Air calibration and solution calibration Salinity compensation (0.0% to 5.0%) Temperature calibration (two points)		
Transmission output hold feature	● Previous value hold ● Optional value hold ● Continuous Selectable from the above (However, only the previous value hold is available in the maintenance mode.)		
Self-diagnosis function	Calibration failure	Zero error and sensitivity error	
	Sensor diagnosis	Short-circuit or disconnection of temperature sensor	
	Measuring range error		
	Converter error		
Temperature-compensated range	0°C to 40°C		
Ambient temperature	-5°C to 45°C		
Relative humidity	20% to 85% (without dew condensation)		
Storage temperature	-25°C to 65°C		
Power supply	Rated voltage 100 V AC to 240 V AC, 50/60 Hz, 10 VA (max)		
Structure	Indoor-use panel installation type Panel case: ABS, Terminal: PBT Panel: IP65 dust and water proof structure		
Protective structure	Panel: IP65 (IEC60529, JIS C0920) Rear case: IP20, Terminal:IP00 Class II device (IEC61010-1) Pollution level 2 (IEC61010-1)		
Conforming standards	CE Marking	EMC Directives (89/336/EEC) EN61326:1997/A1:1998 /A2:2001 Low Voltage Directive (73/23/EEC) EN61010-1:2001	Immunity (industrial environment) Noise increase ±0.4 mg/L or less
	FCC Rule	FCC Part15	

External dimensions	48(W) mm x 96 (H) mm x 115 (D) mm Case depth: approx. 105 mm		
Mass	Approx. 400 g		
Compatible sensors	Probe	DP-100	Operating temperature: 0°C to 45°C
	Sensor	5400, 5405	

7.3 Saturated dissolved oxygen for temperatures and salinities (mg/L)

Temperature	Salinity (%)										
(°C)	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
0	14.16	13.74	13.32	12.90	12.48	12.06	11.64	11.22	10.80	10.38	9.96
1	13.77	13.37	12.96	12.55	12.14	11.73	11.33	10.92	10.51	10.10	9.70
2	13.40	13.01	12.61	12.22	11.82	11.42	11.03	10.63	10.24	9.84	9.45
3	13.05	12.66	12.28	11.89	11.51	11.13	10.74	10.36	9.98	9.59	9.21
4	12.70	12.33	11.96	11.59	11.21	10.84	10.47	10.10	9.72	9.35	8.98
5	12.37	12.01	11.65	11.29	10.93	10.57	10.21	9.85	9.48	9.12	8.76
6	12.06	11.71	11.36	11.01	10.66	10.31	9.96	9.61	9.26	8.91	8.55
7	11.76	11.42	11.08	10.74	10.40	10.06	9.72	9.38	9.04	8.70	8.36
8	11.47	11.14	10.81	10.48	10.15	9.82	9.49	9.16	8.83	8.50	8.17
9	11.19	10.87	10.55	10.23	9.91	9.59	9.27	8.95	8.63	8.31	7.99
10	10.92	10.61	10.30	9.99	9.68	9.37	9.06	8.75	8.44	8.12	7.81
11	10.67	10.37	10.07	9.76	9.46	9.16	8.86	8.55	8.25	7.95	7.65
12	10.43	10.13	9.84	9.55	9.25	8.96	8.67	8.37	8.08	7.78	7.49
13	10.20	9.91	9.63	9.34	9.05	8.77	8.48	8.20	7.91	7.63	7.34
14	9.98	9.70	9.42	9.14	8.86	8.59	8.31	8.03	7.75	7.47	7.20
15	9.76	9.49	9.22	8.95	8.68	8.41	8.14	7.87	7.60	7.33	7.06
16	9.56	9.30	9.04	8.77	8.51	8.24	7.98	7.72	7.45	7.19	6.93
17	9.37	9.11	8.86	8.60	8.34	8.09	7.83	7.57	7.31	7.06	6.80
18	9.19	8.94	8.68	8.43	8.18	7.93	7.68	7.43	7.18	6.93	6.68
19	9.01	8.77	8.52	8.28	8.03	7.79	7.54	7.30	7.05	6.81	6.56
20	8.84	8.60	8.37	8.13	7.89	7.65	7.41	7.17	6.93	6.69	6.45
21	8.68	8.45	8.22	7.98	7.75	7.51	7.28	7.05	6.81	6.58	6.34
22	8.53	8.30	8.07	7.84	7.61	7.39	7.16	6.93	6.70	6.47	6.24
23	8.39	8.16	7.94	7.71	7.49	7.26	7.04	6.81	6.59	6.36	6.14
24	8.25	8.03	7.81	7.58	7.36	7.14	6.92	6.70	6.48	6.26	6.04
25	8.12	7.90	7.68	7.46	7.25	7.03	6.81	6.59	6.38	6.16	5.94
26	7.99	7.77	7.56	7.35	7.13	6.92	6.70	6.49	6.28	6.06	5.85
27	7.87	7.66	7.44	7.23	7.02	6.81	6.60	6.39	6.18	5.97	5.75
28	7.75	7.54	7.33	7.12	6.92	6.71	6.50	6.29	6.08	5.87	5.66
29	7.64	7.43	7.23	7.02	6.81	6.61	6.40	6.19	5.99	5.78	5.57
30	7.53	7.33	7.12	6.92	6.71	6.51	6.30	6.10	5.89	5.69	5.48
31	7.43	7.22	7.02	6.82	6.61	6.41	6.21	6.00	5.80	5.60	5.39
32	7.33	7.12	6.92	6.72	6.52	6.31	6.11	5.91	5.71	5.50	5.30
33	7.23	7.03	6.82	6.62	6.42	6.22	6.02	5.82	5.61	5.41	5.21
34	7.13	6.93	6.73	6.53	6.33	6.13	5.92	5.72	5.52	5.32	5.12
35	7.04	6.84	6.64	6.44	6.23	6.03	5.83	5.63	5.43	5.23	5.02
36	6.95	6.75	6.55	6.34	6.14	5.94	5.74	5.54	5.33	5.13	4.93
37	6.86	6.66	6.45	6.25	6.05	5.85	5.64	5.44	5.24	5.03	4.83
38	6.77	6.57	6.36	6.16	5.96	5.75	5.55	5.34	5.14	4.93	4.73
39	6.68	6.48	6.27	6.07	5.86	5.66	5.45	5.24	5.04	4.83	4.63
40	6.60	6.39	6.18	5.97	5.77	5.56	5.35	5.14	4.94	4.73	4.52

7.4 Parts list

7.4.1 Options

Part names	Models	Specifications	Part codes
Holder	DH-10-1.0	Immersion type holder length (1 m standard)	9220100500
	DH-10-1.5	Immersion type holder length (1.5 m)	9220100600
	DH-10-2.0	Immersion type holder length (2 m)	9220100700
Mounting bracket	SDK-1		9220101500
Mounting bracket	DN-50	SUS	9220101400
Relay box	CT-20DO		9220110000
Junction cable	C-7E	Processed to the specified length (40 m or less)	9100101700
Probe	DP-100	Immersion type (throw-in type) PVC titanium CR	9220102300

7.4.2 Consumables

Part names	Models	Specifications	Part codes
DO sensor	5400	Operating temperature range : 0°C to 40°C Operating pressure range : 0 MPa to 0.1 MPa Material : FEP Film thickness : 25 µm	9220540000
	5405	Operating temperature range : 0°C to 40°C Operating pressure range : 0 MPa to 0.1 MPa Material : FEP Film thickness : 50 µm	9220540500
Consumables that come with the product			
Mounting bracket	-	Parts that come with the product	9010900100
Terminal cover	-		9010900200
Gasket	-		9010900300

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